

Design and Development of a Telepresence Robot for the Covid-19 Pandemic

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ABSTRACT

In the following document we will discuss telepresence robots and give our ideas on how we can make one to help in the fight against covid-19. We will cover the history of the robots, how they have been applied in the world for covid-19. Then discuss the problem we have found to be an issue with robots and how then we can solve the selected problem with initial ideas.

Keywords

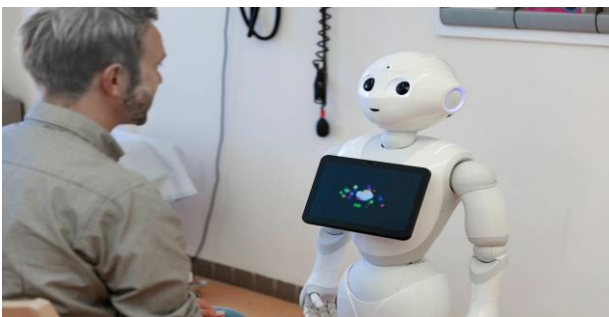
Covid-19, Telepresence Robot, Engineering Design, Product Development, CAD, SolidWorks, Simulation

1. INTRODUCTION

The Covid-19 pandemic has created an unforgettable worldwide health catastrophe, necessitating the development of creative methods to lessen its far-reaching impacts. Telepresence robots are one such option that has become a technology darling. These adaptable robotic gadgets have proven their usefulness in a variety of fields, with the biggest two applications in the healthcare and education field. They offer connectivity and remote presence when physical presence is constrained. The broad range of applications for telepresence robots and their impact on society during the epidemic are examined in this research segment.

2. WP1- BACKGROUND AND LITERATURE REVIEWS

2.1 TASK 1.1: BACKGROUND AND LITERATURE REVIEW ABOUT TELEPRESENCE ROBOTS



A telepresence robot (TPR) is a motorized apparatus that can be moved and is controlled remotely. It has several interactive features that make it ideal for remote cooperation and communication. Marvin Minsky coined the word "telepresence" in 1980, and it refers to the sensation of being fully present at a remote site from one's own physical location, producing a virtual or simulated environment for the robot operator (Rae et al., 2015).

Sensor-assisted mobility control, a screen, speakers, microphones, cameras, and other interactive features are commonly found on telepresence robots. When the robot is positioned on a screen, distant users can control its movements using their computers, tablets, or cell phones while seeing the surrounding area. Simultaneously, the user's face is projected live onto the screen of the robot, allowing for face-to-face conversation with onsite workers and establishing a sensation of remote telepresence (Perez, 2020).

Commercially available telepresence robots can do things like change the robot's height to match the user's height, move the robot's head, and provide advanced capabilities like real-time full resolution zoom for high-quality visual communication. Some models may also include laser pointers, auto-navigation, and mapping. However, these robots' capabilities for manipulating items in remote areas and providing tactile feedback are still limited (Davey, 2021).

What are the main applications of telepresence robots?

Telepresence robots' principal applications include supporting older persons in maintaining their independence, offering remote visits and communication, and improving the projection capability of secondary users (Cesta et al, 2016). They can be utilised in small flats where the elderly dwell to interact with their surroundings and receive support from remote operators. Telepresence robots can also be utilized for remote visits and communication, where the robot serves as an operator's proxy, making them appear to be in a different place (Cesta et al, 2016). Moreover, the robot can consistently and safely navigate its surroundings by using its autonomous skills, which allow it to locate itself, map its surroundings, and avoid impediments. These applications were inspired by the extensive field testing of the Giraff robot. (Cesta et al., 2016).

Integration into the workplace, remote communication, social contact, and establishing long-distance partnerships are among the primary uses of telepresence robots. Individuals who are unable to physically be present in a certain place can nonetheless communicate with each other thanks to telepresence robots. According to Kristoffersson et al. (2018), they empower remote workers to feel present and mobile at work, making it easier to be ad hoc available and included in unplanned dialogues. Telepresence robots in schools offer comparable advantages by letting pupils participate in class discussions and eat lunch with their classmates. Even for small-scale social interactions during academic conferences, telepresence robots have been demonstrated to be advantageous. Telepresence robots provide a way to connect people in long-distance relationships by simulating some aspects of face-to-face interactions (Kristoffersson et al., 2018). It is important to note that the context given does not encompass all potential uses for telepresence robots; additional uses may exist that are not covered here.

What are the technical and economic challenges for Telepresence Robots?

Handling telepresence robots is one of the technological challenges they present, and it can be especially difficult during installation and user interaction. Robots that are not working well can cause users distress, like disconnections. Upkeep, including battery longevity, is another problem. Furthermore, patients and caregivers using the robot need to have an online connection. The literature review of studies on the use of telepresence robots with older individuals, their families, and professional caregivers brought these difficulties to light (Isabet et al., 2021).

Telepresence robots face several technological difficulties, such as the labour-intensive nature of driving, their limited expressiveness in terms of body language, their limited self-awareness about appearance and sound levels, and their limited field of vision. The usage of telepresence robots in a variety of settings presents additional privacy issues. Improvements including expansive or panoramic views, movable heights, noise-sensitive audio controls, and privacy-preserving features have all been recommended by researchers. On the other hand, not much study has been done on using telepresence robots in the home to facilitate communication between family members. Telepresence robots' direct access may be useful, but it also jeopardizes privacy, claim Yang & Neustaedter (2018). A few design ideas that could be used to solve these problems are availability modes, rule-setting, and notifications of incoming connections. Furthermore, the possibility of placing two telepresence robots in different locations adds to the cost and design challenges. Generally speaking, design opportunities exist to examine the simultaneous connection of two sites while taking into account users' attachment to the devices (Yang & Neustaedter, 2018).

One of the financial obstacles facing telepresence robots is the robot's acquisition and upkeep costs. Concerns over the affordability of purchasing a robot for personal use were voiced by both professionals and individuals. When evaluating Telepresence Robots against other non-robotic communication technologies, it is crucial to take their cost into account. Examining the financial and human expenses related to using a robot also necessitates performing a cost analysis. It also stresses training users, particularly professionals and family members, how to operate the robot and navigate its interface. Overall, these financial difficulties show how important it is to assess the costs and resource usage of telepresence robots (Isabet, 2021).



In June 2022, 190 results from the domains of computer science (50%), robotics (39%), engineering (28,4%), automation control systems (6.8%), and education and educational research (6.3%) were returned in the Web of Science database when the keyword "telepresence robots" was entered in the topic area. Since 2006, when the database's first TPR publication was added, the quantity of publications has gradually increased. China (5.8%), Japan (12.1%), Canada (7.9%), Australia (6.8%), and the United States (37.4%) make up the majority of the donations.

Almost all professional or commercial contexts use TPRs (telepresence robot), or may in the future (Reis et al., 2018). Paulos and Canny (1998) investigated the PROP (Personal ROving Presence) system, which is the subject of the first study on the phenomenon of mobile robotic telepresence. The use of TPRs in a range of contexts and fields, including medicine (Daruwalla et al., 2010), interpersonal communication (Ogawa et al., 2011), museums (Roberts & Arnold, 2012), and education (Kwon et al., 2010; Tsui et al., 2011; Tanaka et al., 2014; Conti et al., 2017), has since been the subject of numerous studies.

Marvin Minsky, one of the pioneers in this field of study and cofounder of the Artificial Intelligence Laboratory at the Massachusetts Institute of Technology (MIT), first used the term "telepresence" in 1980 to describe teleoperation systems that allow users to manipulate remote physical objects (Rae et al., 2015). The sensation of being totally present at a distance from one's own physical place is known as telepresence. For the robot operator, telepresence generates a virtual or simulated environment of the real experience (ElGayar et al., 2005). A TPR is a moveable, wheeled, remote-controlled gadget with cameras, microphones, speakers, a screen, sensor-assisted motion control, and other interactive capabilities specifically made for remote collaboration and communication. Through the use of laptops, tablets, or smartphones, a remote user can access the robot and control it while viewing the onsite environment on a screen. Concurrently, the user's face is being live projected into the robot's screen, enabling onsite staff to interact with the user face-to-face and creating a remote telepresence. (Perez, 2020)

2.2.1 Application fields of telepresence robots

1-Healthcare Applications:

Hospital robots are working diligently to support humanity during the COVID-19 pandemic. Robots are not just sterilising rooms and carrying supplies and medication, but they are also enabling hospital workers to work much more safely and effectively through telepresence.

Telepresence robots have been implemented in hospitals, allowing nurses to visit patients without worrying about wearing personal protective equipment, and allowing doctors who are isolated at home to remain present in emergency rooms. By providing a little extra flexibility to an overworked system, the robots are reducing the danger of infection while enabling hospital staff to remain productive whether they are at home or in quarantine themselves. (Ackerman, 2023)

2- Education and Remote Learning:

The COVID-19 pandemic has resulted in school shutdowns, causing huge problems for everyone concerned, including parents, teachers, and children. The problem of this situation is that missing courses can have a big impact on grades and even the motivation to keep studying.

The telepresence robot assists in solving the issue. The robot in school can be accessed by a pupil at home. Subsequently, the robot assumes the role of their "avatar," capable of attending classes, taking notes during lectures, viewing experiments during chemistry and physics lessons, responding to questions from the instructor, and taking examinations. Students can snap pictures of their written assignments and forward them right away to the teacher for review. (Carl Cullinane and Rebecca Montacute, 2020), (Victor Denisenko, November 27, 2021)

Challenges of using Telepresence Robots

One of the biggest challenges is poor resources and infrastructure. Numerous users experienced poor connection quality, which resulted in call disruptions between the interviewer and the resident. (Wilson et al., 2022)

The novelty of telepresence robots and the interviewers', residents', and medical staff's unfamiliarity with the technology posed another hurdle to the use of these devices for interviews; at first, several researchers were hesitant to employ them due to their novelty. Furthermore, because interviewers, residents, or staff are more accustomed to these techniques, they might have preferred doing interviews over the phone or virtually via Zoom or other platforms. Research was hampered in this instance by the novelty and unfamiliarity with telepresence robots. (Hung et al., 2023)

Table 1- Examples of most popular Telepresence Robots

		
(a) Beam	(b) Giraffe	(c) Double 3

Fig. 1 Example Telepresence Robots

a) Beam

Small company owners can drive to a coworker's desk, grandparents are able to speak with their grandchildren, parents can read bedtime stories to their kids or sing to them, and people can even just check in on their pets while they are away from home. "Party Mode" is among Beam's exciting new features. The Beam+ can automatically adjust to louder surroundings thanks to an audio function called party mode. Both the speaker volume and the integrated noise reduction are amplified, allowing for improved sound quality regardless of the surroundings. Another innovative feature to look out for is the option to add limbs and other attachments that are still in development.

Cost: \$2,140.00

Available at- <https://telepresencerobots.com/robots/suitable-technologies-beam/>

b) Giraff

Seniors who are unable to physically visit their friends and family can maintain connections with them thanks to the Giraff telepresence robot. The robot may be manoeuvred to observe objects nearby, virtually as if you were there in person, and the picture of the remote user is shown on the screen.

Price: \$11,900.00

Available at- <https://telepresencerobots.com/robots/giraff-telepresence/>

c) Double 3

Double 3 is a two-wheeled, self-driving videoconferencing robot that is revolutionising remote work and education. Double provides a physical presence to those who are unable to be in person, allowing telecommuters, remote professionals, and students to feel more connected to their colleagues. You may go about the workplace, attend meetings, visit other worksites, and take classes from anywhere in the world when you have your own Double.

Price: £5,000-7,000

Available at- <https://robosavvy.co.uk/double-3-full-set.html>

Available at- <https://www.doublerobotics.com/>

Conclusion

In brief, telepresence robot applications have aided senior citizens who are unable to see one another due to visiting restrictions, which has led to severe communication issues both during and after the COVID-19 pandemic. Furthermore, telepresence robots can be used safely and easily to check on patients from home or other secure locations due to a shortage of medical staff.

3. WP2- DEVELOPMENT OF PRODUCT DESIGN SPECIFICATION AND CONCEPT DESIGN

3.1 TASK 2.1: DESIGN PROBLEM, DESIGN BRIEF AND DEVELOPMENT OF PRODUCT DESIGN SPECIFICATION.

3.1.1 Design Problem and Design Brief

In the following project we have been given a task to utilize a telepresence robot primarily for covid application. So, we will need to innovate and make a new kind of telepresence robot, it must be able to withstand being in a hospital environment during the covid period. Within the hospital the robot should be able to help and assist with any situations the robot is presented with for example helping patients and their health and able to do some nurses/doctor roles without a human being needed in the room to limit interaction and time around the infected patient. We have decided to go with a budget of £250,000 for the robot with the aim of once R&D the machines specifically for the use for producing these robots is to be able to sell them for £2,500 and still make a profit. With a timeframe of one year to produce at least a beta version of a working robot but preferably a fully functional robot with no issues such as bugs in the coding or mechanical errors. The main challenge we will have to overcome is how the operator would be able to use and control the robot accurately to meet the patients' needs in a variety of different ways. Another challenge would be how it is able to help slow the spread of covid.

In the sections above it has been covered how there are many different approaches this project can be taken, with the use of the previously made robots as inspiration. But by looking at the previously designed robots and the problems they've faced, we will have to design a robot that is able to have a secure and reliable connection between the robot and the operator, as well as to help make the robot more affordable as the other made robots are considered to be very expensive so another focus is on the price and making it affordable.

Key stake holders would be the following; staff at the hospital, the operator, patients, family and friends of the patient, the manufacture, the investors and possible shareholders of the investing company, possible sponsorships, the competition and the public that use the hospital.

3.1.2 Development of the product design specifications

Camera and screen

The robot is required to have a camera and screen which can be used to allow visual communication from the operator and the people that are local to the robot when in operation. The camera can be further utilised by having a 360-degree camera which can be accomplished with one or multiple cameras.

The screen also needs to be able to be easily seen and large enough where the people in proximity can easily see the operator and what they are trying to display.

Speakers and microphone

To allow communication between local people relative to the robot and the operator in the terms of audio, a speaker and microphone need to be used. The microphone needs to be able to withstand the various interference from audio around the robot which can be accomplished in terms of software and or hardware.

The speakers can have multiple different uses so larger and louder speakers would be preferred to be installed but are not necessary.

Strength

The robot can be attacked or damaged in the medical field as people might be intimidated by the robot and people might attack the robot out of anger. It needs to withstand some impact and damage to the body and parts.

Mobility

Due to the robot being in an area which can be unknown what is on the ground the mobility options are needed to be specific. Due to that factor the robot is preferred to have the ability to traverse over many kinds of terrain that can be achieved in multiple ways. But if tires are used it is preferred to use airless tires to help with chance of failure while in operation. But all movability of the robot must be powered by electricity such as an onboard battery.

Communication/ encryption

The robot will need to have long range communication either directly long range, to a cell tower or equivalence. But the onboard software will need to encrypt the outgoing signal to help with security. As well as inbound signals will need to be decrypted.

Location

To help with operation and using the robot there is a requirement to have location tracking abilities on the robot to be able to track and record the operation of the robot. The location of the robot must still be available after malfunction and damage to the robot to help locate others around the location.

Storage

The robot must have various size storages available to be able to equip it with various options to help utilise its use and help with it with efficiency in the field. The largest equipment must be enough size to fit a large person in (2032.00 mm) (Statistics normal distribution, accessed 23/10/23, nd) and the smallest storage available size must be around the size of 20mm by 50mm by 150mm.

Medical

The robot must also have the arms and features available to it to apply basic medical aid to civilians as well as the equipment to do so for example gauss and bandages as a few examples. As well as having a feature to give medicine in the form of some but preferred all the following: liquids, tablets, injections, and sprays. As well as having a thermometer to be able to detect the temperature of the patient as an example of some of the medical equipment the robot will be needed, another can be a sphygmomanometer.

Cleaning/Hygiene

Due to the large risk of spreading of covid within a hospital environment the robot can help keep the area clean. This can be done in many ways such as using a UV light or chemicals. With the robot being powered with a battery though a UV light can be very useful and then won't leave any dangerous materials behind keeping the patients and staff safer.

Application/In use

The operator must be able to;

- Control the movement and direction of the robot.
- Be able to turn speakers/microphone on and off.
- Be able to turn the screen/camera on and off.
- Open and close all storage spaces.
- Some storage spaces must be able to be moved, for example having medical equipment in a storage space to move the item/s closer to the person or in particular orientation.
- Full control of the arms.
- Being able to give the medicine needed.
- Having use of some basic medical equipment.
- Able to run a self-diagnostic to find any problems with the robot.
- If chosen to have moveable armour, then to be able to move those as well.

Table 2- Level of importance for the design specification needs

No	Need	Interpretation of the need	Importance 1-5
1	<u>Camera and screen</u>	The robot is required to have a camera and screen which can be used to allow visual communication from the operator and the people that are local to the robot when in operation. The camera can be further utilised by having a 360-degree camera which can be accomplished with one or multiple cameras. The screen also needs to be able to be easily seen and large enough where the people in proximity can easily see the operator and what they are trying to display.	5
2	<u>Speakers and microphone</u>	To allow communication between local people relative to the robot and the operator in the terms of audio, a speaker and microphone need to be used. The microphone needs to be able to withstand the various interference from audio around the robot which can be accomplished in terms of software and or hardware. The speakers can have multiple different uses so larger and louder speakers would be preferred to be installed but are not necessary.	5
3	<u>Strength</u>	The robot can be attacked or damaged in the medical field as people might be intimidated by the robot and people might attack the robot out of anger. It needs to withstand some impact and damage to the body and parts.	4

4	<u>Mobility</u>	Due to the robot being in a warzone the terrain can be varied. Due to that factor the robot is preferred to have the ability to traverse over many kinds of terrain that can be achieved in multiple ways. But if tires are used it is preferred to use airless tires to help with chance of failure while in operation. But all movability of the robot must be powered by electricity such as an onboard battery.	4
5	<u>Communication/encryption</u>	The robot will need to have long range communication either directly long range, to a cell tower or equivalence. But the onboard software will need to encrypt the outgoing signal to help with security. As well as inbound signals will need to be decrypted.	4
6	<u>Location</u>	To help with operation and using the robot there is a requirement to have location tracking abilities on the robot to be able to track and record the operation of the robot. The location of the robot must still be available after malfunction and damage to the robot to help locate others around the location.	3
7	<u>Storage</u>	The robot must have various size storages available to be able to equip it with various options to help utilise its use and help with it with efficiency in the field. The largest equipment must be enough size to fit a large person in (2032.00 mm) (Statistics normal distribution, accessed 23/10/23, nd) and the smallest storage available size must be around the size of 20mm by 50mm by 150mm.	5
8	<u>Medical</u>	The robot must also have the arms and features available to it to apply basic medical aid to civilians as well as the equipment to do so for example gauss and bandages as a few examples. As well as having a feature to give medicine in the form of some but preferred all the following: liquids, tablets, injections, and sprays. As well as having a thermometer to be able to detect the temperature of the patient as an example of some of the medical equipment the robot will be needed. Another can be a sphygmomanometer.	5
9	<u>Cleaning/Hygiene</u>	Due to the large risk of spreading of covid within a hospital environment the robot can help keep the area clean. This can be done in many ways such as using a UV light or chemicals. With the robot being powered with a battery though a UV light can be very useful and then won't leave any dangerous materials behind keeping the patients and staff safer.	4

Fig. 2 (Level of importance for the design specification needs)

Table 3- Ideal values for the needs

No	Metric	Specification	Units	Marginal value	Ideal value
1	1A 1B	Camera degree of view Screen	Degrees Yes/No	180-360 N/A	360 Yes
2	2A 2B	Speakers Microphone	Yes/No Yes/No	N/A N/A	Yes Yes
3A	3AA	Blunt force protection	Newtons	2000-3000	2500(Boxing Science, 2014)
4	4A 4B	Powered by electricity. Airless tires.	Yes/No Yes/No	N/A N/A	Yes Yes
5	5A 5B	Encryption of signal out. Description of signal in.	Yes/No Yes/No	N/A N/A	Yes Yes
6	6A	Location/GPS.	Yes/No	N/A	Yes
7	7A 7B 7C	Large storage. Small storage. Various storage size.	MM MM Yes/No	Height 1500-2500 10-30 by 35-65 by 100-200 Yes/No	Height 2032.00 mm 20mm by 50mm by 150mm. Yes
8	8A 8B 8C 8D	Medicine. Arms. Apply basic medical aid. Basic medical equipment.	Yes/No Yes/No Yes/No Yes/No	N/A N/A N/A N/A	Yes Yes Yes Yes
9	9A	Cleaning/Hygiene	Yes/No	N/A	Yes

Fig. 3 Ideal values for the needs

3.2 TASK 2.2: CONCEPT DESIGN -DEVELOPMENT OF DESIGN CONCEPTS AND SELECTION OF THE BEST DESIGN CONCEPT.

Function Analysis

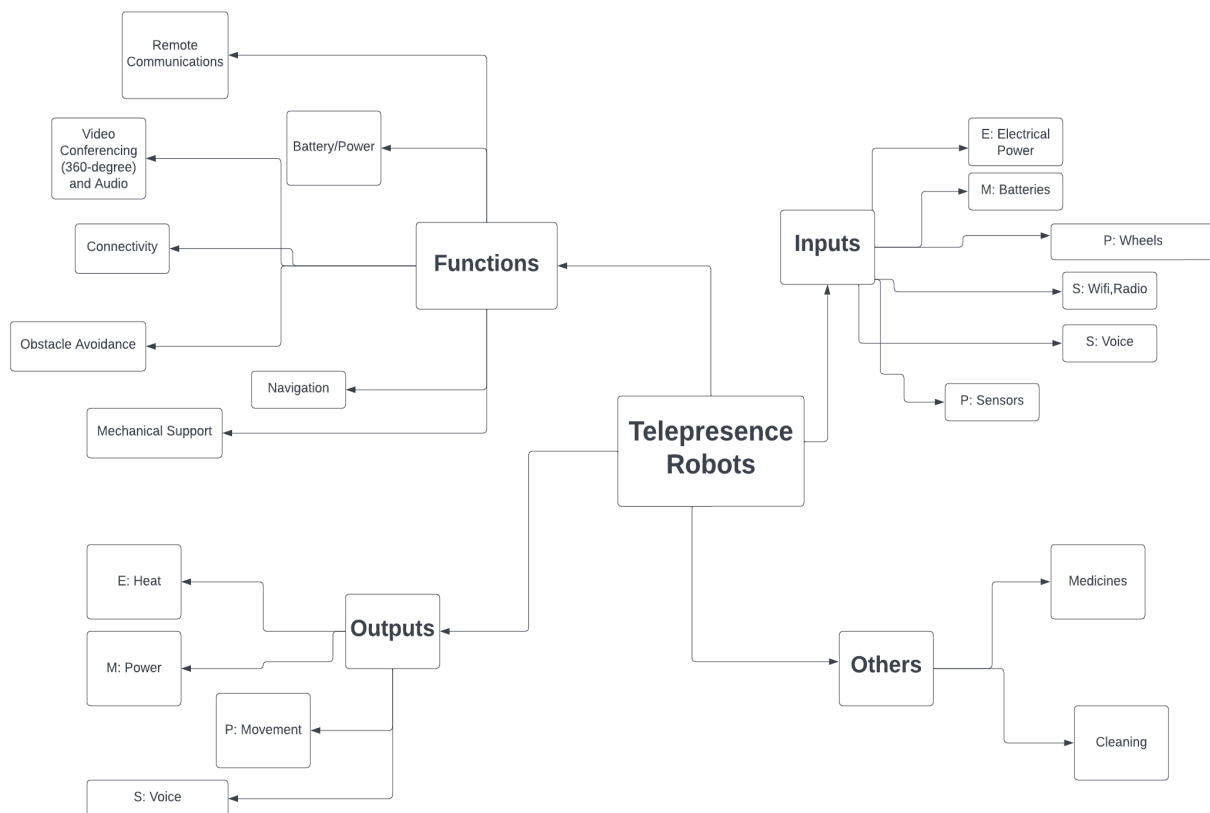


Fig. 4 Function Analysis Diagram

Function diagram:

The primary inputs for our TPR are the physical environment, its inventory of medical supplies, telecommunications signals, and local or remote user input. It must be capable of navigating its environment, physically interacting with environmental objects and users, transporting its inventory and a user, and communicating via telecommunications or audiovisual output.

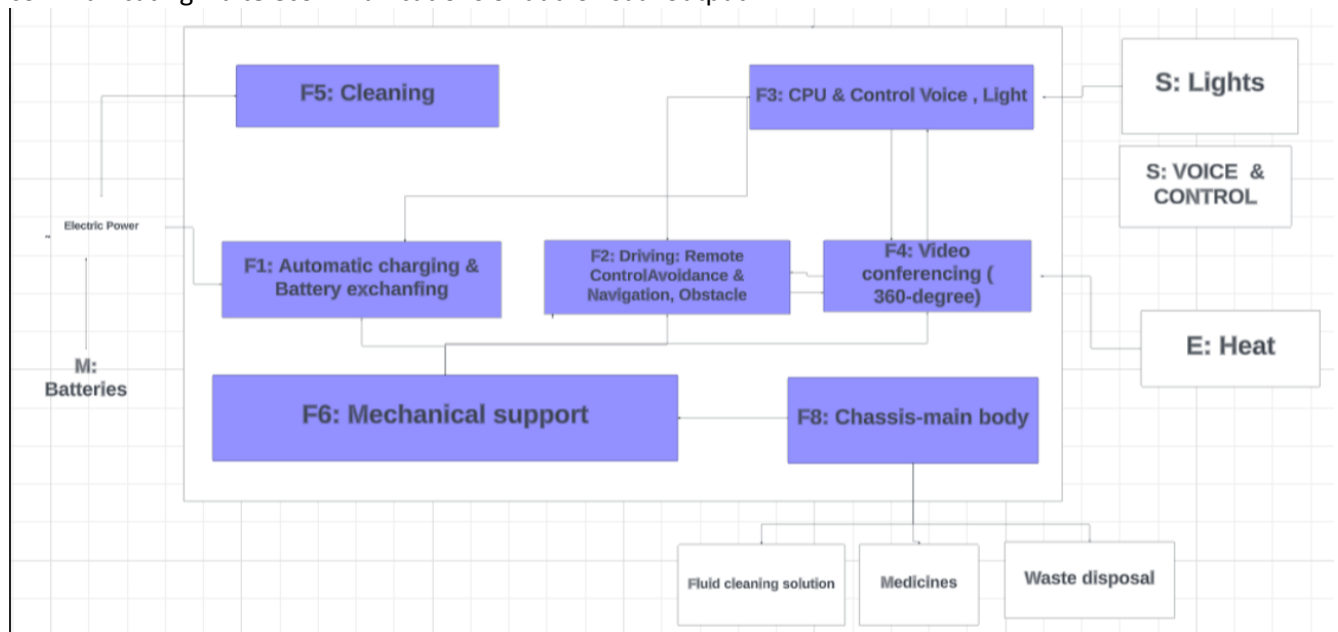
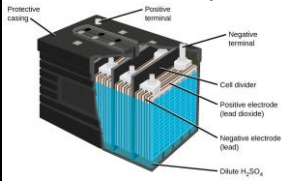


Fig. 5 Function Diagram

By using the above diagram, you can see the various inputs and what functions they are linked with. The main input would be the batteries which will allow for all functions of the robot to be working such as running the CPU and internal processes, with the robot's movement using electricity being a very important part of the robot. With the various other inputs, the robot can do the relevant function.

Another function would be to detect if the batteries need to be charged and monitored. Needing the cleaning function can be done in multiple ways in which we have covered such as the UV light or the brush which can help limit the spread of the virus. With the controls from the operator, the CPU will be used to control the robot's movement and navigation of the robot. Another function which is linked with the movement would be its 360-degree camera to view its surroundings. The mechanical support will be for self-diagnostics to detect and try to solve the problem but if not then it can create a log for an operator to then fix. The last function is for the main body to have multiple functions such as the dispensing of the medicine and other items to the patient and staff. Another function of the main body is to store chemicals and other things for the robot function such as the optional chemical cleaning that the robot will use to help kill covid and other micro-organisms. The main body can also be used to take and then dispose of any waste from the robot or the patients.

Morphological Chart

Function	Solution 1	Solution 2	Solution 3
Remote Communication 	Voice Controller 	Mobile App Controller 	Motion-activated switches 
Power/Battery 	Lead sulfate 	Li-ion 	Fuel Cell + Battery 
Connectivity 	Wifi 	Satellite 	Cellular 

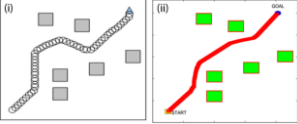
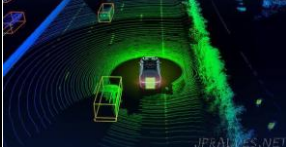
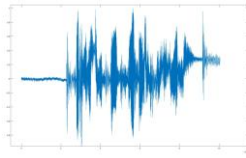
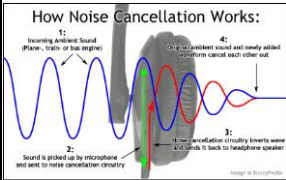
Video Conferencing 	Single Camera 	360-degree camera 	PTZ Camera 
Obstacle Avoidance 	Stereo Vision Cameras 	Ultrasonic Sensors 	LiDAR sensors 
Navigation 	Tracks 	Wheels 	Legs 
Audio Transmission 	Multi-directional mic 	Uni-directional mic 	Noise Cancellation 

Fig. 6 Morphological Chart

Sketch Concepts:

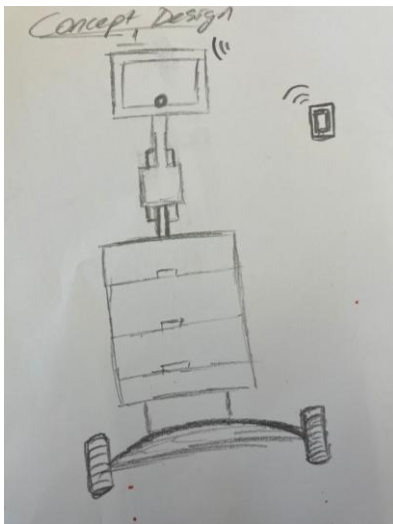


Fig. 7 Concept Design 1

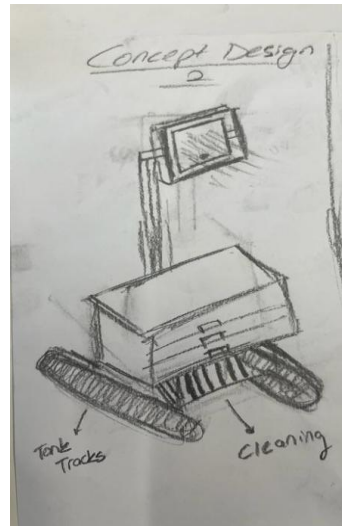


Fig. 8 Concept Design 2



9 Concept Design 3

Several sketch concepts were generated using combinations of features from the morphological chart. For concept design 1, we selected wheels for navigation and highlighted the presence of a remote communication device, drawing from our morphological chart for key functional features to display. For the second concept, we experimented with an alternative storage form-factor and a more rugged drive system utilizing a tracked design, which would allow for a heavier lead-acid battery. Finally, the third concept displays how a unidirectional microphone could be mounted and demonstrates our third drive system option in the form of bipedal locomotion (presented as “legs” in the morphological chart).

Rendered concept designs from Vizcom using our sketches (Vizcom AI (Artificial Intelligence) [Online], accessed 2023)



A- Fig. 10 Rendered Concept A



B- Fig. 11 Rendered Concept B



Fig. 12 Rendered Concept C

Using the Viscom tool, several rendered design concepts were generated from initial sketches emphasizing the established core features. Render A features a wheeled design, with the cleaning systems clearly shown at the base of the robot and draws in the main body for storage. Render B uses tracks for mobility, allowing a more stable design at the cost of a less humanoid form-factor, necessitating a long strut above the body to position the video screen at head height. Render C moves in the opposite direction, with a more humanoid form achieved at the cost of stability and some cleaning functionality through the use of actuated leg struts.

3.2.1 Selections of the best design concept or design solution

Concept Screening:

Criteria	Reference	Concept 1	Concept 2	Concept 3
Communication	0	0	-1	+1
Mobility	0	0	-1	-1
Durability	0	+1	+1	-1
Cost	0	0	0	-1
Total	0	+1	-1	-2
Rank		1	2	3
Continue?		YES	YES	NO

Fig. 13 Concept Screening Table

To screen our design concepts, the Giraff TPR was used as a reference competitor product. Then, we selected several key criteria from the specification and compared whether each of our designs performed better or worse than the reference product in those areas, before discarding the lowest-scoring design.

Concept Scoring:

		Concept 1		Concept 2	
Criteria	Weight	Rating (1-5)	Score	Rating (1-5)	Score
Communication	30%	4	1.2	3	0.9
Mobility	25%	3	0.75	2	0.5
Durability	25%	2	0.5	4	1
Cost	20%	3	0.6	2	0.4
Total Score			3.05		2.8
Continue?		YES		NO	

Fig. 14 Concept Scoring Table

To select a design for final development, we used the same set of key criteria and analysed in more detail how important each was to our product, and how well our designs performed in those areas. Then, we used those weights in combination with our performance ratings to give each design a final score and selected the highest-scoring concept for further development.

Concept Selection:

The final concept selected for development was Design Concept 1, the wheeled design. This is because the form-factor of the robot allows for the best balance between a humanoid design and one that is mobile, stable, and easy to manufacture. As well as this, the design best supports key features unique to our product such as the combination of cleaning and storage capabilities on a single device.

4. WP3- DEMONSTRATIONS OF EMBODIMENT AND DETAIL DESIGN

4.1 TASK 3.1 - 3D CAD MODELLING AND DESIGN

In order to determine the general proportions of the robot that we would need to utilize as a guide while building our SolidWorks versions of it, Luke and Mohamed spoke and drew some rough sketches of the robot seen in the above picture. There are additional sketches in the appendix.

Our team successfully created the telepresence robot in a visually attractive SolidWorks file by carefully adhering to the specified dimensions and size parameters. This digital depiction demonstrates the accuracy attained during the design process by accurately reflecting the dimensions shown in the previously given picture. Our CAD platform of choice, SolidWorks, was essential to our project since it provided a wealth of capabilities and methods that not only fulfilled but also beyond our design goals.

A wide range of tools and methods were used during the SolidWorks design process, including as cutting, extrusion, drawing, helix generation, mirroring, and more. With the use of these technologies, we were able to move from rough 2D sketches to a finished 3D model, demonstrating the effectiveness and adaptability of computer-aided design (CAD). Interestingly, SolidWorks makes it possible to create 3D models that are useful for more than just viewing; they are an essential tool for ideation and design refinement prior to actual manufacturing.

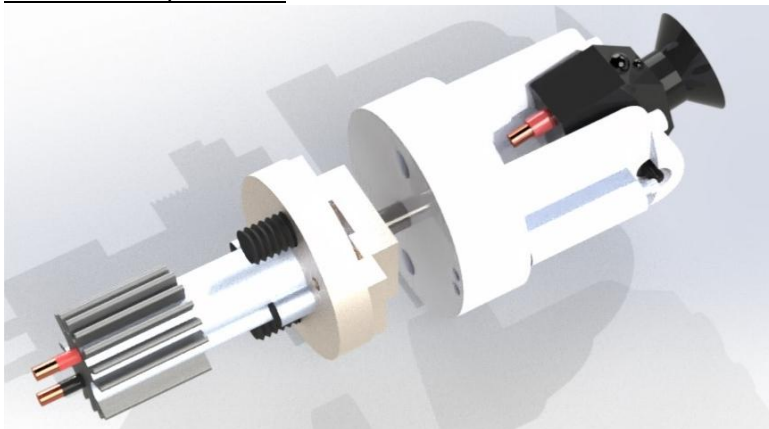
Among the best CAD programs, SolidWorks is notable for its extensive features. It is possible to generate machine code, which is an essential part of the manufacturing process, in addition to visual models.

SolidWorks code is able to be easily incorporated into CNC machining procedures, directing the equipment to precisely cut, drill, or shape materials. This feature guarantees the physical reproduction of SolidWorks-designed components with accuracy, which facilitates the smooth transition from digital design to real prototypes or production units.

SolidWorks is much more versatile in current manufacturing since it can be used to produce items using 3D printing in addition to CNC machining. The manufacturing and prototyping processes can be streamlined by immediately translating SolidWorks 3D models into printable formats. SolidWorks' versatility makes it an indispensable tool for both visualizing ideas and integrating them smoothly into a range of production processes.

We were able to effectively construct complex and symmetrical structures thanks to the combination of capabilities included in SolidWorks, such as helix generation and mirroring. These characteristics were crucial in realizing the intended telepresence robot design, together with the precise tools included into SolidWorks. A few samples of the 3D model are shown below to give you an idea of our design process. An assortment of more renderings is available in this document's appendix for a more thorough examination. These images demonstrate the superior design quality produced by SolidWorks and highlight the crucial role it plays in bridging the conception and physical reality gaps in the field of robotic production.

Camera on top of robot



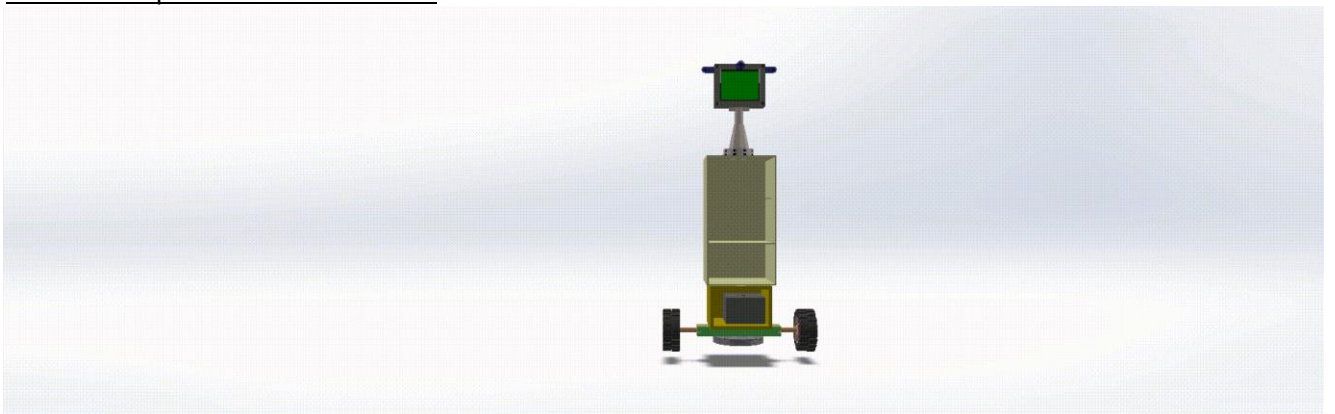
Wheel



Exploded view animation of the wheel.



Full robot exploded view animation.



(Adobe MP4 to gif)

To better visualize the robot and its components—which can even be exhibited in an animation—more SolidWorks capabilities, such as exploded views, were employed. This helps to make the pieces easier to comprehend visually and how they are put together. Moreover, animations were employed to assist in demonstrating the robot's movement. Another aspect of SolidWorks is its ability to simulate how a robot would behave in the real world with real-world forces and physics applied to it.

Bill of materials for the fully built robot.

Item number	Part	Qty	Description	Manufacturing method	Material
1	Front frame of screen	1	Provides structural support and outer casing for the display panel.	Injection molding	High-strength plastic or aluminum alloy
2	25% of tv Frame	4	Contributes to the overall structure of the TV, providing stability and support.	Extrusion or CNC machining	Aluminum or steel
3	Back of the frame	1	Encloses internal components and provides protective casing.	Injection molding or die casting	ABS plastic or aluminum
4	TV	1	Main body housing the display, electronics, and other components.	Assembly of various parts	Plastic and metal components
5	screws for tv frame	4	Secures different parts of the TV frame, ensuring structural integrity.	Machining or cold forming	Stainless steel
6	screws for front to the back of tv	4	Secures the front frame to the back frame.	Machining or cold forming	Stainless steel
7	speakers	1	Audio output components providing sound for the TV.	Assembly of speaker components	Speaker cone (paper, plastic, or metal), housing (plastic or wood)
8	Motor	8	Used for various functions such as adjusting screen orientation.	Precision manufacturing	Copper coils, magnets, and steel
9	camera holder	4	Provides support and stability for cameras within the TV.	CNC machining or injection molding	Lightweight and durable plastic or aluminum alloy
10	second motor 0.5	4	Smaller motors for specific functions within the TV.	Precision manufacturing	Copper coils, magnets, and steel
11	camera	4	Captures images and videos with protective covers for durability.	Precision assembly with lens elements	Glass or plastic for lenses, durable plastic or metal for the camera body
12	cameras protect	2	Protective covers for cameras.	Precision assembly	Durable plastic or metal
13	main body	1	The primary structure of the telepresence robot, housing essential components.	Assembly of various parts	Lightweight and durable materials
14	middle part	1	Component located at the bottom of the main body, housing CPU and Wi-Fi.	CNC machining or injection molding	aluminum alloy
15	closer of the battery	1	Component securing and enclosing the batteries within the battery chassis.	Assembly of various parts	Plastic or metal
16	battery chassis	1	Housing for the batteries, providing a secure and organized power source.	Injection molding or CNC machining	Durable plastic or aluminum

4.2.1 Selections of components

Based on the existing design, we recognized the need for several prebuilt components. These include a camera, microphone, monitor, speakers, motor, gyroscope, and onboard computer. Each of these was specified according to the table in Appendix: Part Selection. This specification was then used to search a number of robotics supply sites, such as kitronik.co.uk, robotshop.com, and botland.store for parts meeting the requirements. Finally, a qualitative table of their comparative appropriateness for each specified attribute was constructed using the DEXi MCDM tool and, and this was used to sort the parts by overall appropriateness in order to make the best selection for each required part.

Part	Camera	Microphone	Monitor	Speaker	Motor	Gyroscope	Processor
1	LX-OAKDS2	S-DEV-17272	S-LCD-20664	S-COM-14023	UNI-8010-6	S-SEN-21336	S-WRL-16781
2	LX-OAKDPW	S-BOB-18011	S-LCD-15890	S-COM-18343	S-ROB-22132	S-SEN-21325	S-WRL-18573
3	DB-CAMKIT	S-COM-18488	S-LCD-21318	S-DEV-20690	IN-NEMA23	S-SEN-20176	S-DEV-15139
4	S-SEN-14725	S-COM-08635	S-LCD-17153	S-PRT-20660	S-KIT-18625	S-DEV-17270	S-DEV-23551
5	Lx-OAK-DProWPoE	CBOX-Pro	S-LCD-20663	S-COM-11089	S-ROB-14450	DF-SEN0386	S-SPX-21775
6	S-DEV-16260	S-SEN-21219	S-LCD-16774	S-COM-21311	DF-FIT0493	S-GPS-22660	S-DEV-21251
7	S-SEN-21803	S-SEN-19728	S-LCD-20662	S-COM-09151	S-ROB-12779	S-SEN-22857	S-WRL-18594
8	S-SEN-21276	S-DEV-17287	S-LCD-20661	S-BOB-21772	R-902-0017-001	S-KIT-21286	S-DEV-15170
9	LX-A00399	S-DEV-17740	S-DEV-16985	S-COM-15453	S-DEV-14285	S-SEN-19727	S-DEV-21326
10	OB-311	S-BOB-21537	DF-DFR0464	S-BOB-11044	S-ROB-14451	R-903-0257-000	S-DEV-17713

Fig. 15 Components as ranked according to DEXi model

4.2.2 Simulation for design

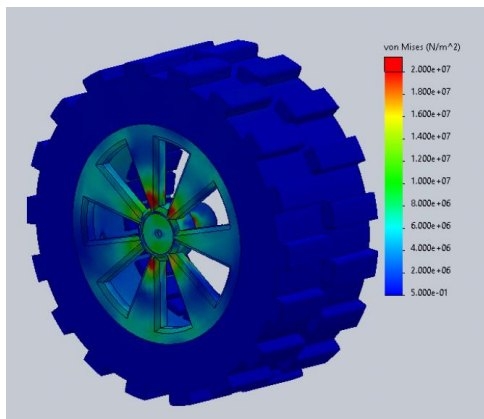


Fig. 16 Stresses on the wheel under load

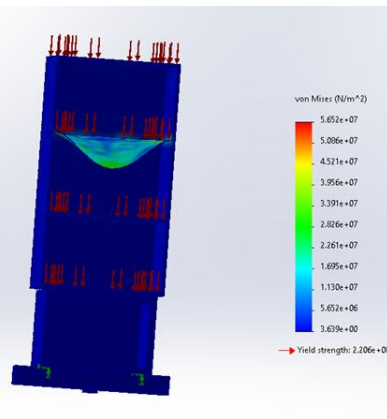


Fig. 17 Shelf Loading Simulation

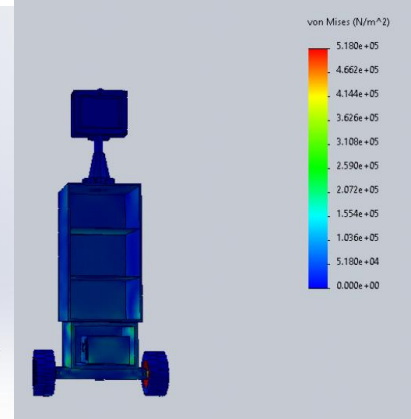


Fig. 18 Side Impact Simulation

Throughout the modelling and detail design process, individual parts and subassemblies were frequently simulated in SolidWorks to test technical suitability of the design with various materials. This informed design decisions, such as the change in spoke shape due to concentrated stresses near the axle as demonstrated in fig.16, a test applying load to the axle of the wheel with a fixed base of the rim in order to test its ability to bear the weight of the robot. It also allowed the team to spot and fix errors such as the inconsistency in shelf width highlighted in fig.17. Finally full-assembly simulation of various scenarios such as shelf overload or impacts, such as in fig.18 (which simulates an impact to the side of the body, such as by a human striking the robot), allowed the team to confirm the fitness-for-purpose of the final design.

4.2.3 Selection of materials

Criterion	Category	Part
Must be easy-to clean	Technical	Body
Must be lightweight and resilient	Technical	Body
Must not exceed 30% of total material and manufacturing cost, including electronics and other pre-built components	Economic	Body
Must not have excessive lead time at low unit manufacturing	Logistics	Body
Must be resilient	Technical	Wheels
Must not exceed 30% of total material and manufacturing cost, including electronics and other pre-built components	Economic	Wheels
Must not have excessive lead time at low unit manufacturing	Logistics	Wheels

Fig. 18 Material requirements

In order to select materials, the MCDM modelling tool DEXi was used. Key criteria (see Fig. 18) were entered and rated for each combination of material and manufacturing method to construct clear visual comparisons and evaluate the overall most appropriate combination.

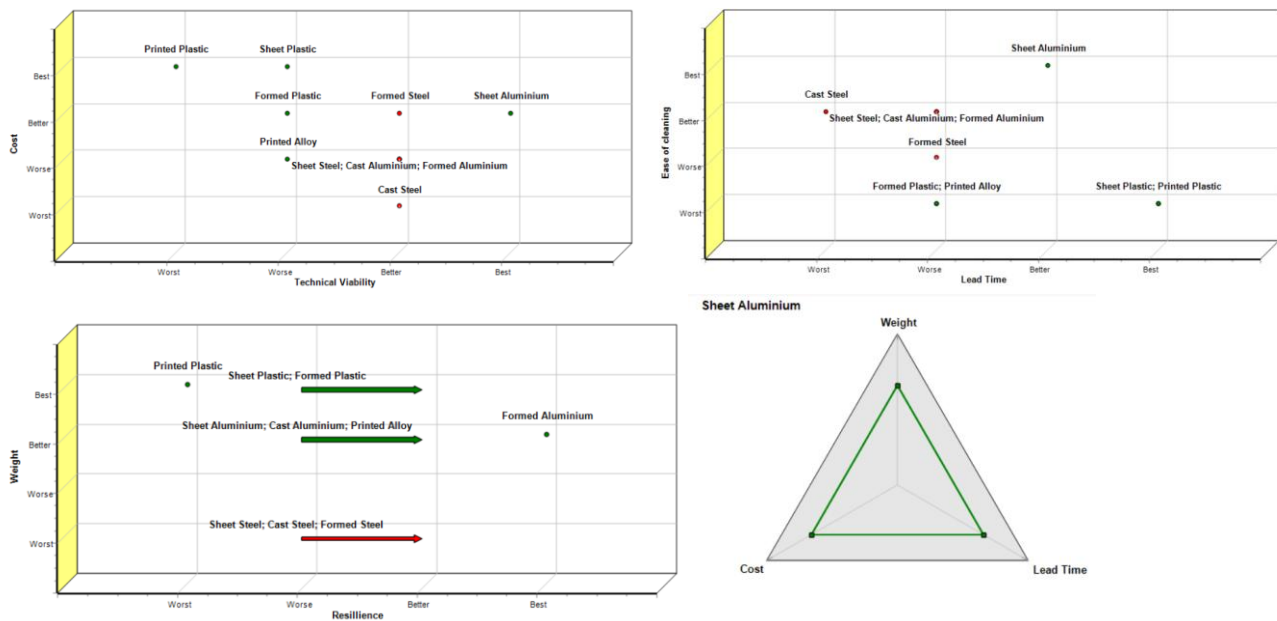


Fig. 19 Multi-Criteria Decision Making

Overall, 5mm 6010 sheet aluminum ordered to size and spot-welded into configuration was selected for the main body due to its low weight, lead time, and manufacturing costs as well as the resilience of the metal and the ease of cleaning nonporous metals (See fig. 19). Meanwhile, the more complex configuration of the wheels requires that they be machined into shape. However, their lower volume as compared to the body brings the cost and lead time for this process into acceptable parameters, while aluminum remains a suitable material due to its resilience and lower cost of manufacture with this process.

5. WP4 - VIRTUAL PROTOTYPING AND DESIGN DOCUMENTATION

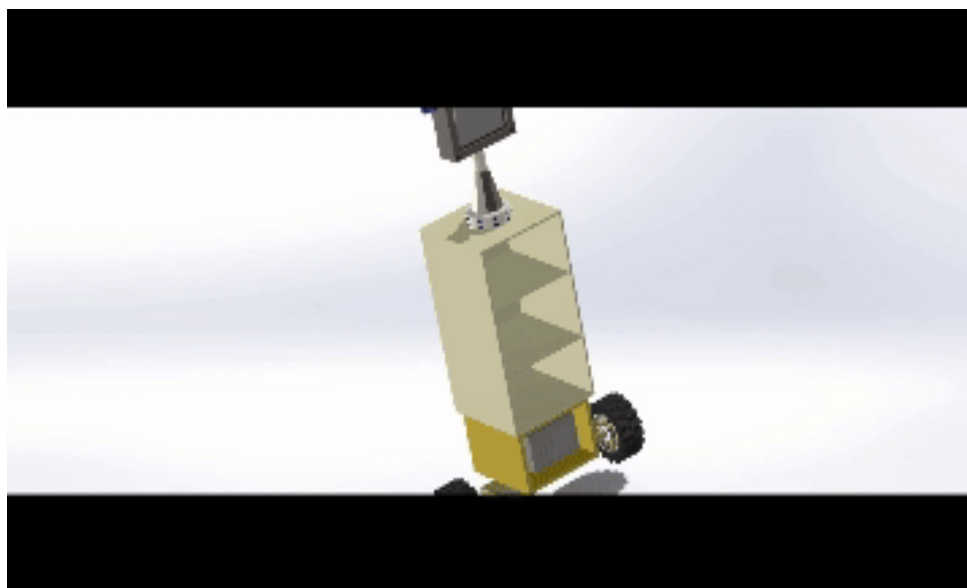
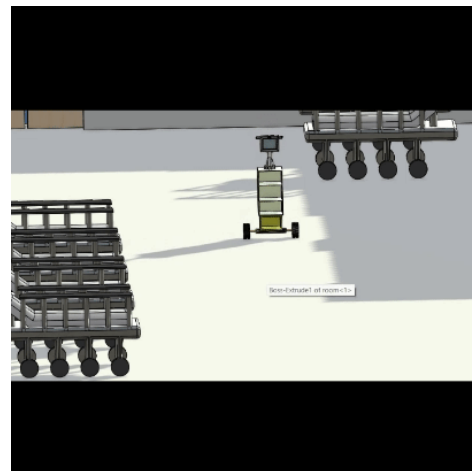
5.1 TASK 4.1 - CREATION OF 3D ANIMATIONS

A telepresence robot's computer-aided design (CAD) animation provides an engaging and educational depiction of a cutting-edge technical tool intended for interaction and communication across long distances. This animation uses cutting-edge CAD software to produce a lifelike and dynamic simulation.

The video starts off by presenting the robot's clean and contemporary look, emphasizing its mobility features—such as wheels and screen—and demonstrating how well it can move across a variety of locations. The intricate elements of the robot's build, such as its sturdy materials and ergonomic layout for maximum performance, are captured in the CAD model.

As the animation goes on, the robot's telepresence powers come into more emphasis. The integration of high-definition cameras, microphones, and speakers is demonstrated using CAD techniques, demonstrating the robot's use as a useful communication tool. The animation shows how users may see, hear, and interact with their environment remotely thanks to the robot's capacity to relay real-time audio-visual data.

Overall, the CAD animation is an effective means of conveying the telepresence robot's operation, design, and possible uses. It offers an engaging visual depiction that may be utilized for informational, instructional, or promotional reasons, assisting stakeholders, prospective users, and the broader public in understanding the cutting-edge features of this technology.



(Adobe MP4 to gif)

5.2 TASK 4.2 - CREATION OF TECHNICAL DRAWINGS

Transforming design thoughts into accurate and thorough illustrations is a complicated process when creating technical drawings for a telepresence robot. Every single element of the robot's design, parts, and functions is captured in the drawings. The drawings include exploded diagrams, isometric views, and orthographic projections, among other perspectives, to give a thorough depiction for assembly and production needs. To achieve precise duplication, consideration is given to elements such as dimensions, tolerance levels, and material requirements. The process of turning creative concepts into real, working robotic systems begins with the development of these sketches. The drawings are in the appendix section.

6. SUMMARY & CONCLUSIONS

Throughout the whole project we have all learnt many new skills and knowledge linked to telepresence robots. With the information gained we understand that there are many aspects which can be improved upon to improve the telepresence robots which cannot all be fitted into one project. With our chosen problems we have practice our new learnt skills from the lectures during the first term such as the use of AI (Artificial Intelligence) to create a visual example of the robot, as well as creating tables to be able to help us focus on the more selected parts of the specification. The information gained from WP 1 and WP 2 it allowed us to present our initial ideas based on the problems discovered such as a connectivity issue between the user and the robot, which we have covered by using multiple different communication ways for example Wi-Fi, Bluetooth and use of cellular connections. With all the information we have gained we have been able to outline a specification and an initial idea of how we are going to meet it. Then we have jointly agreed on the selected robot as the solution to the specification mentioned in the above section.

With this chosen solution we are now able to continue to WP3 and WP4. The chosen solution can now be made into more detail in SolidWorks with more detailed 3D modeling and animations of it in motion. With the size and idea being selected throughout WP1 and WP2 the budget can be now used as reference to get the ideal material and dimensions to be made. After doing that step with using the specification and function table as reference the robot can then be analyzed and optimized to be able to meet the specification more accurately. With it being then improved and specifically made in detail the technical drawings then can be made which then can be used to create the robot.

7. ACKNOWLEDGMENTS

Our thanks to Faculty of Engineering and Science, University of Greenwich for the support. Academic and technical staff at University of Greenwich are acknowledged for advice and guides during the time of project implementation.

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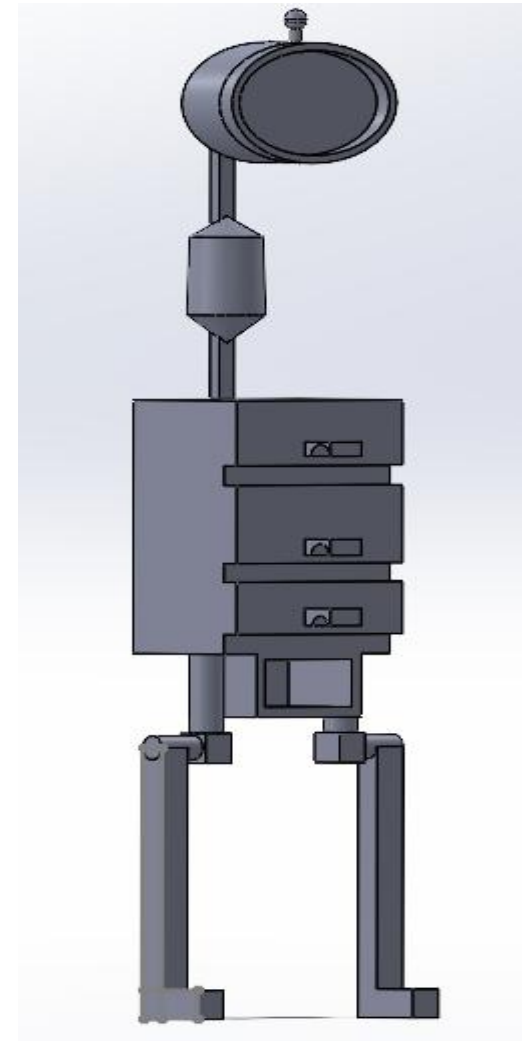
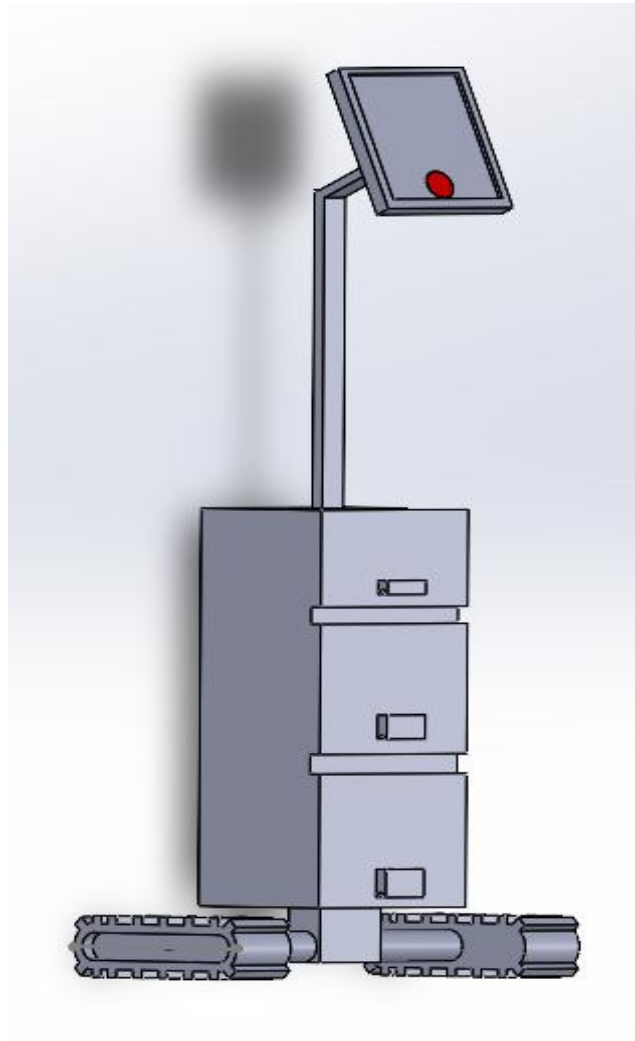
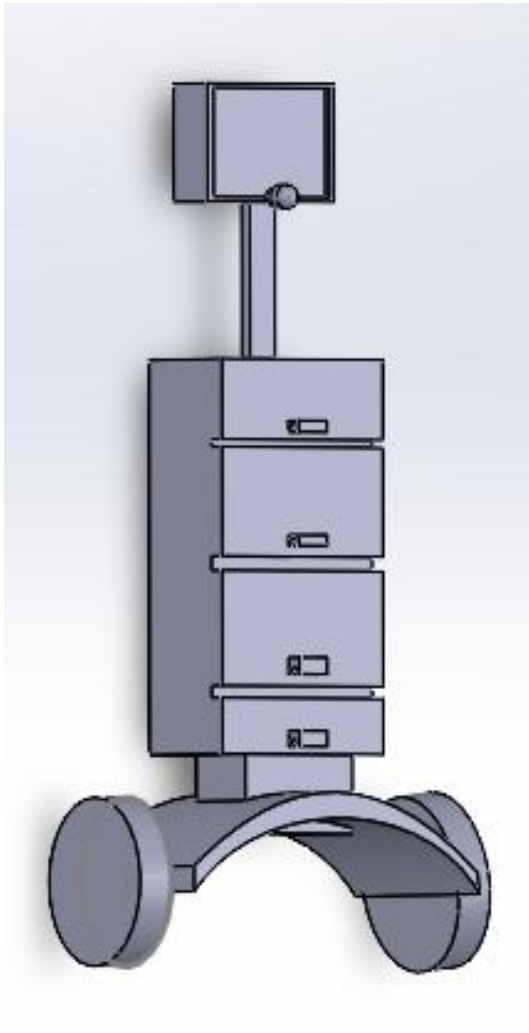
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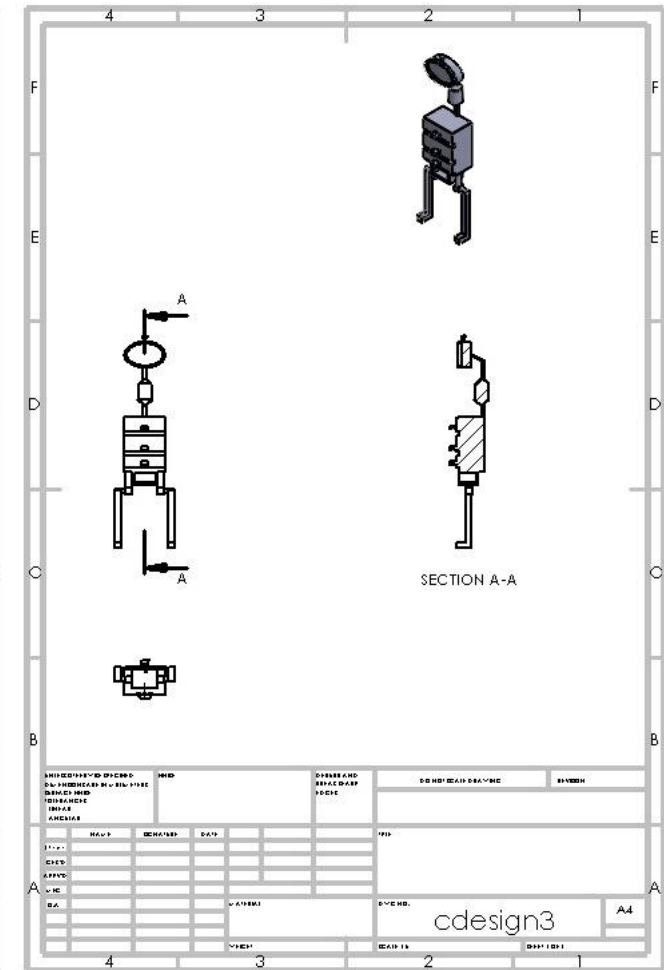
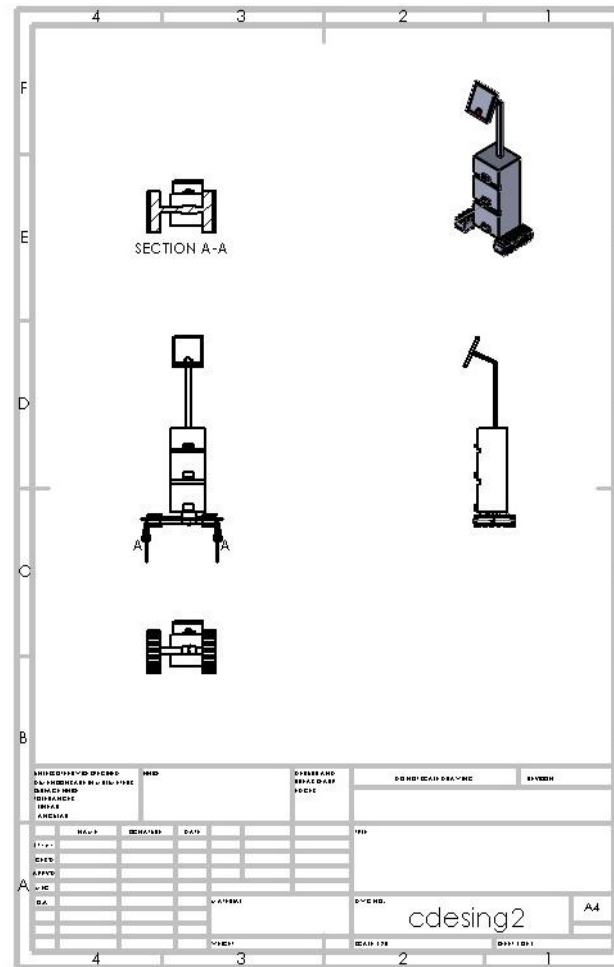
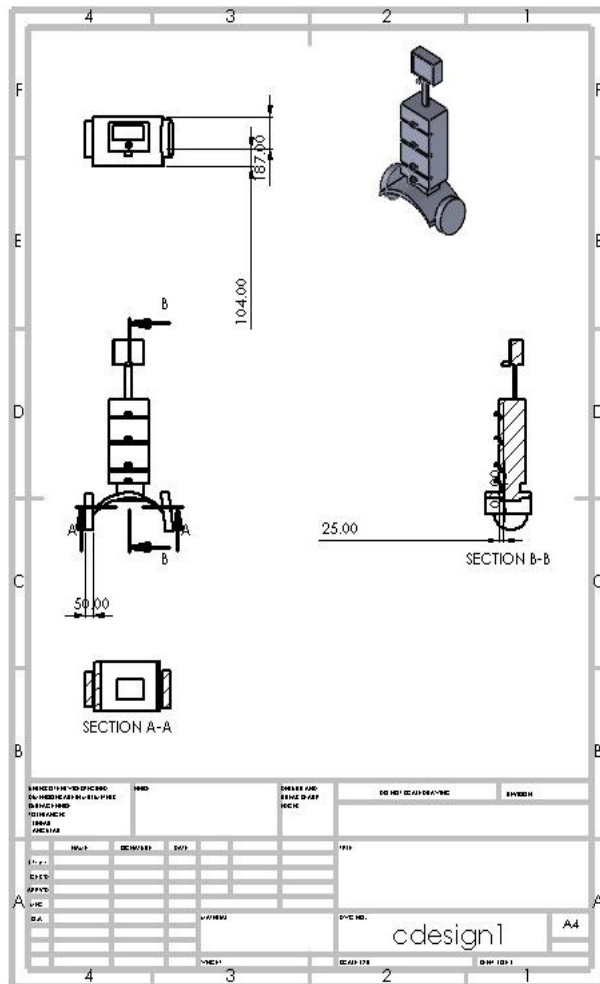
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SAMPLE APPENDIX TO PRESENT FIGURES ABOUT 3D CAD DESIGN OF TELEPRESENCE ROBOTS

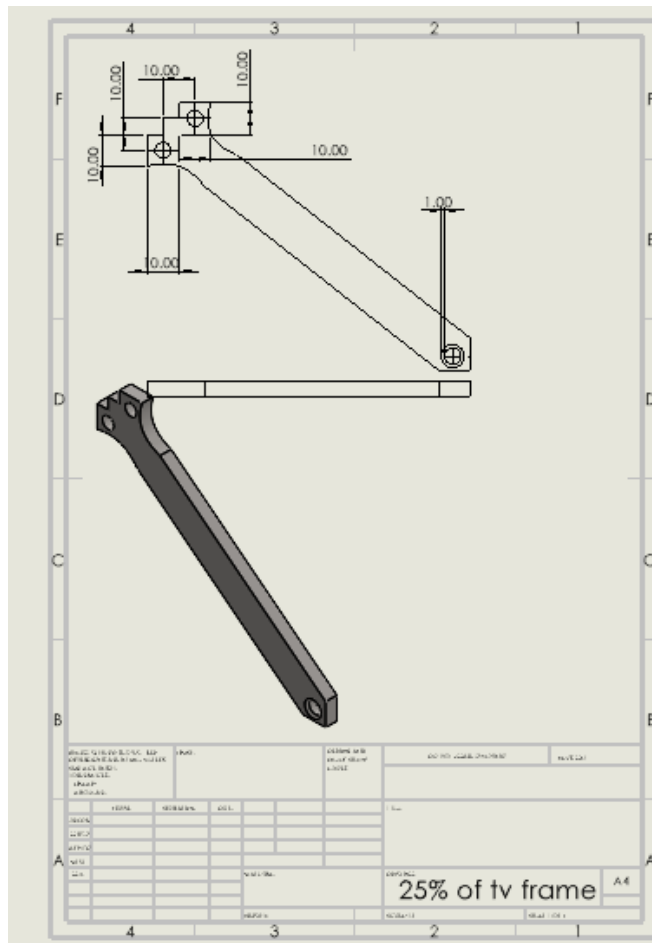


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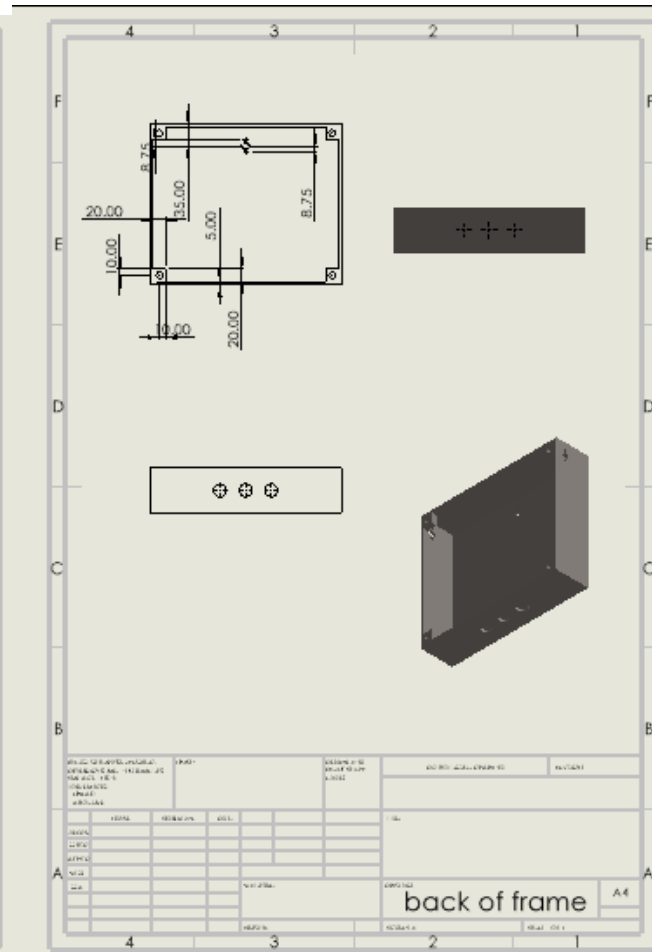


Technical drawings

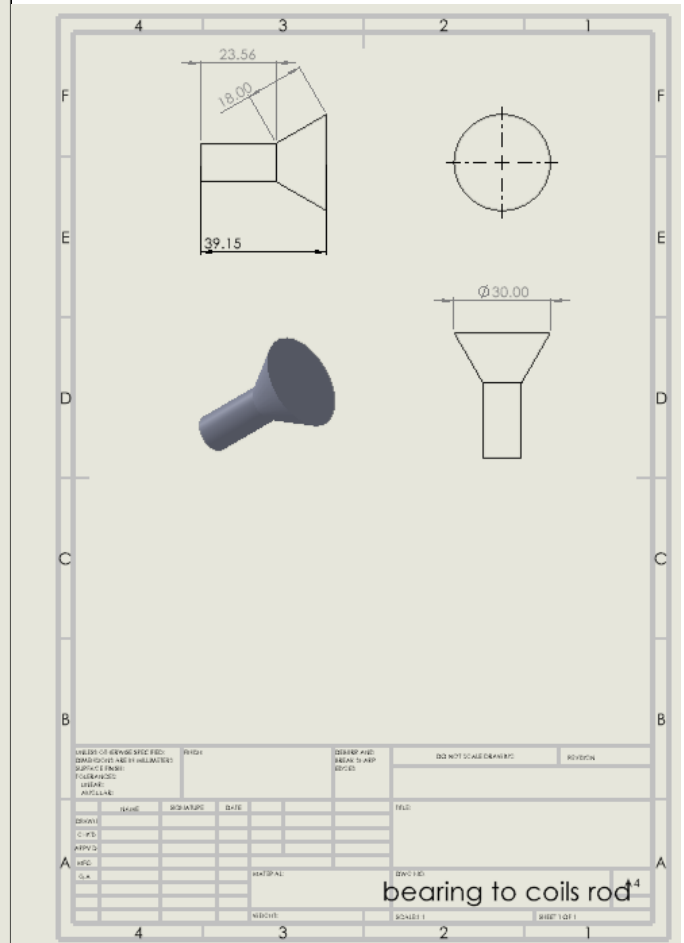
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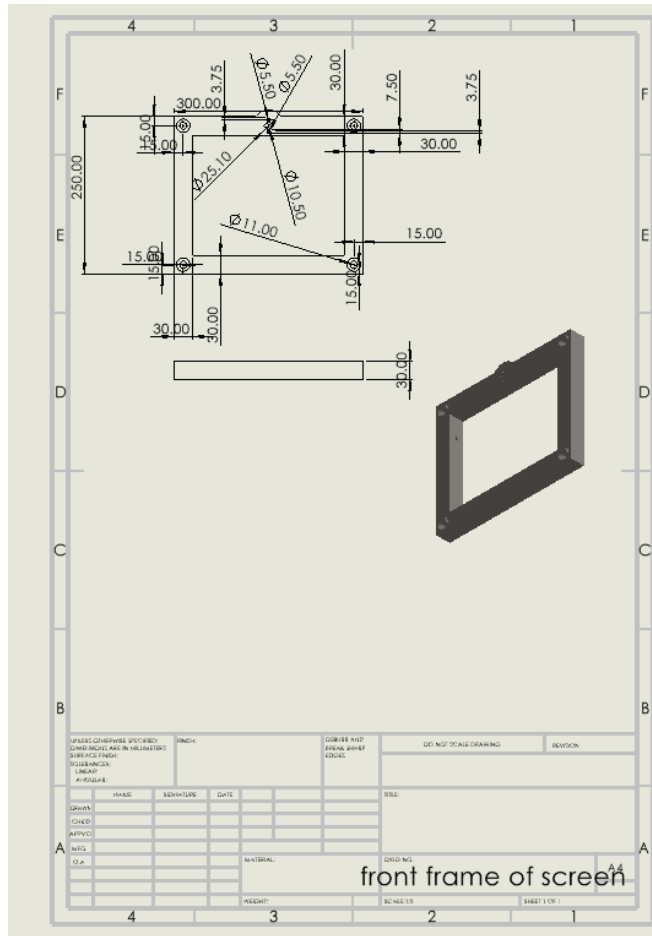
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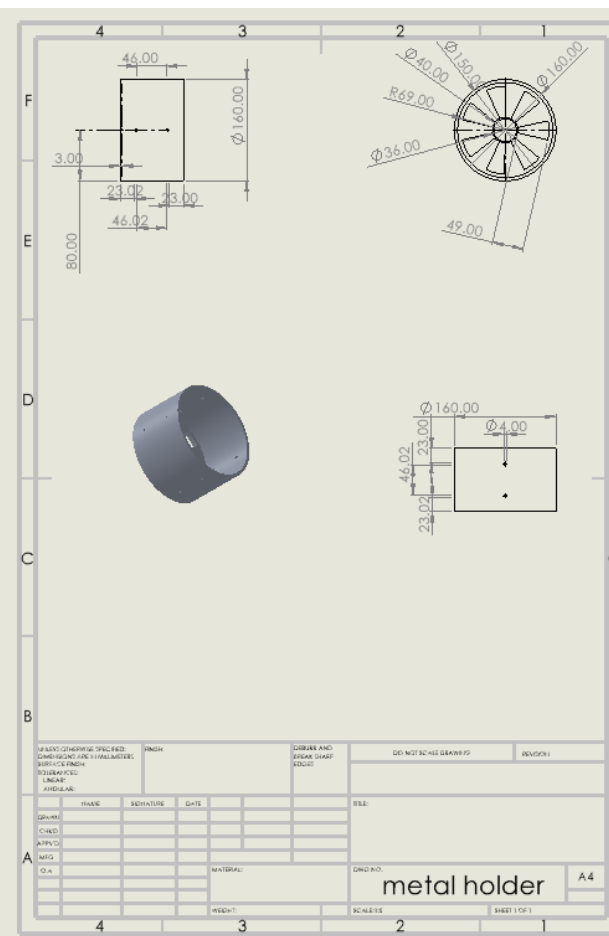
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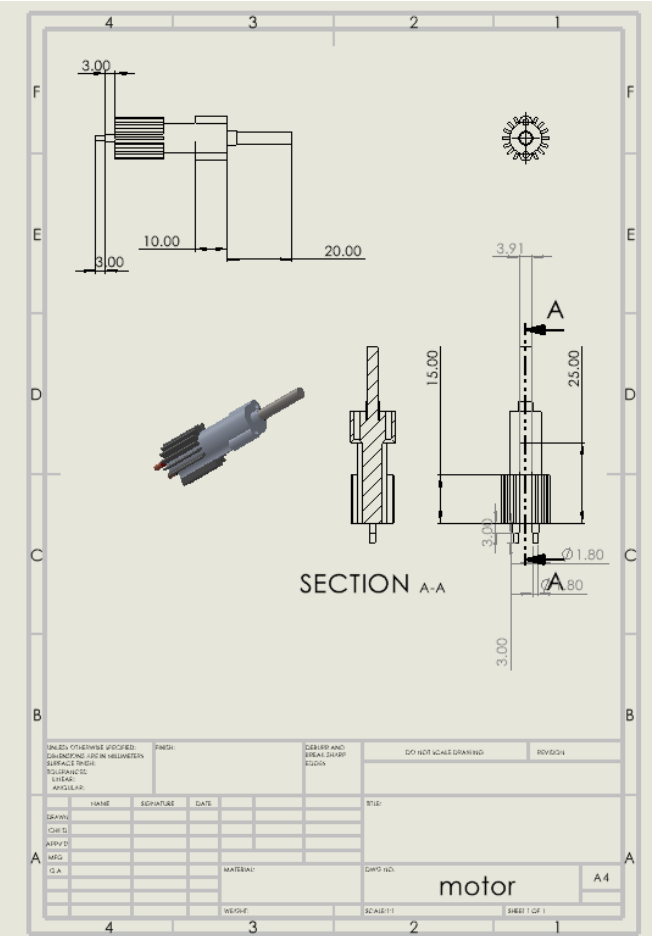
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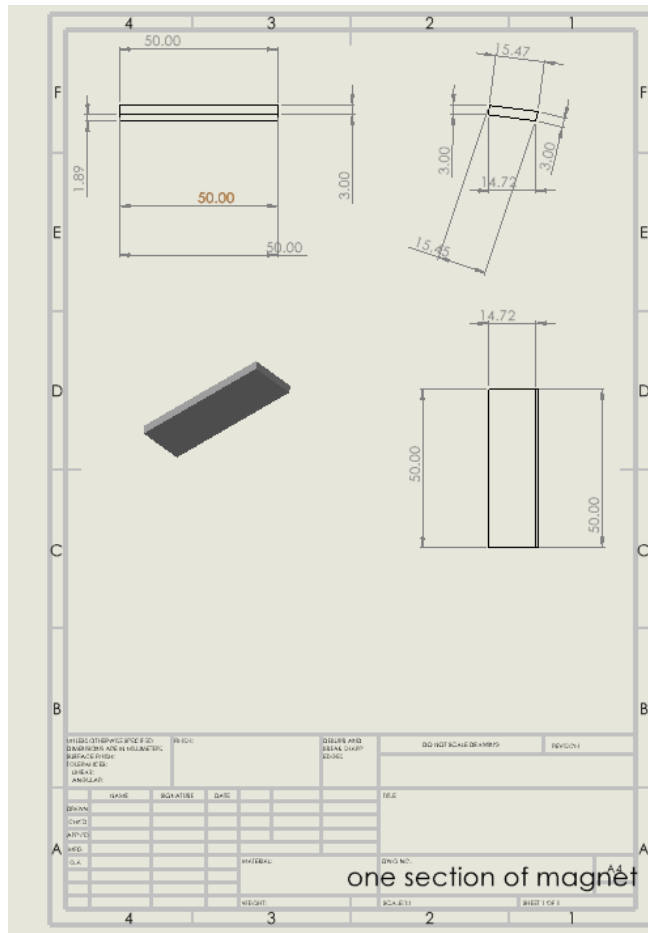
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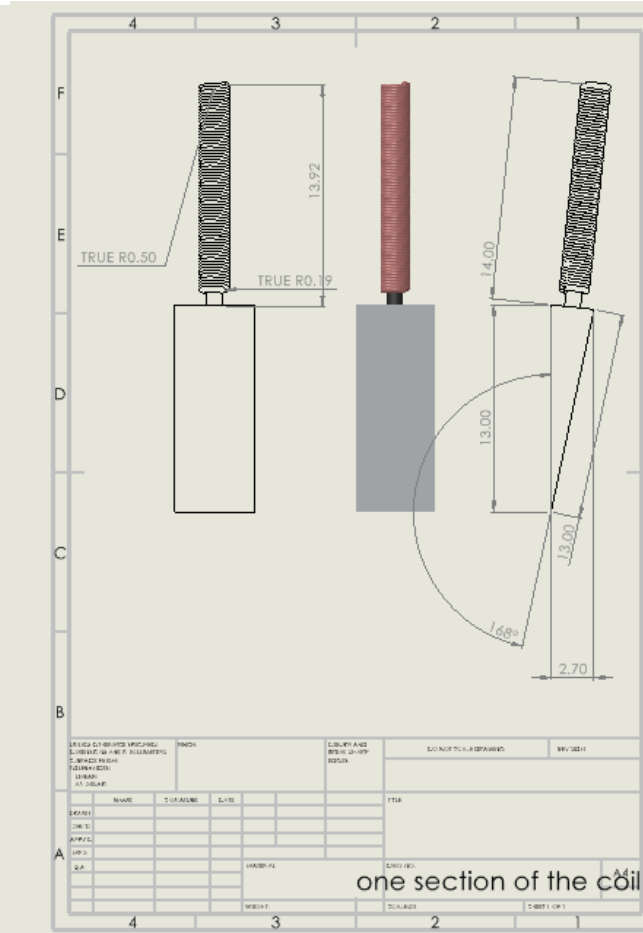
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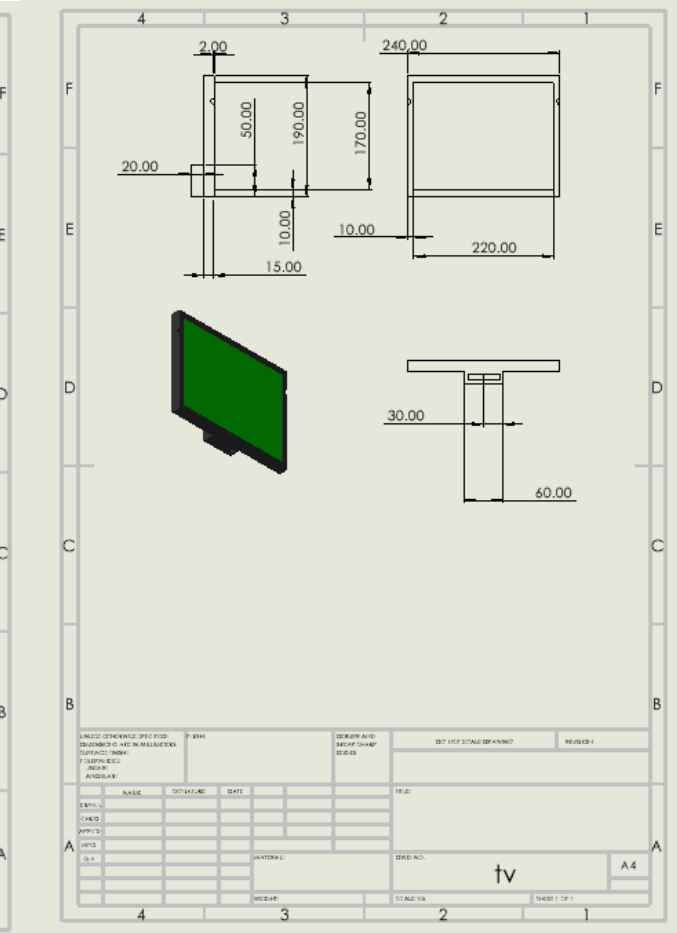
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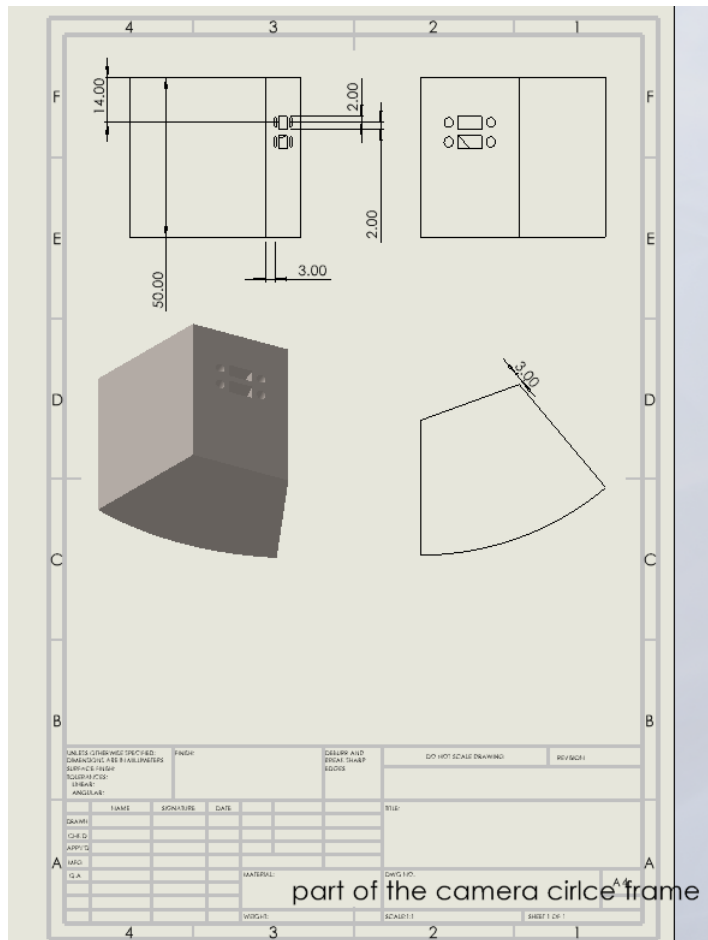
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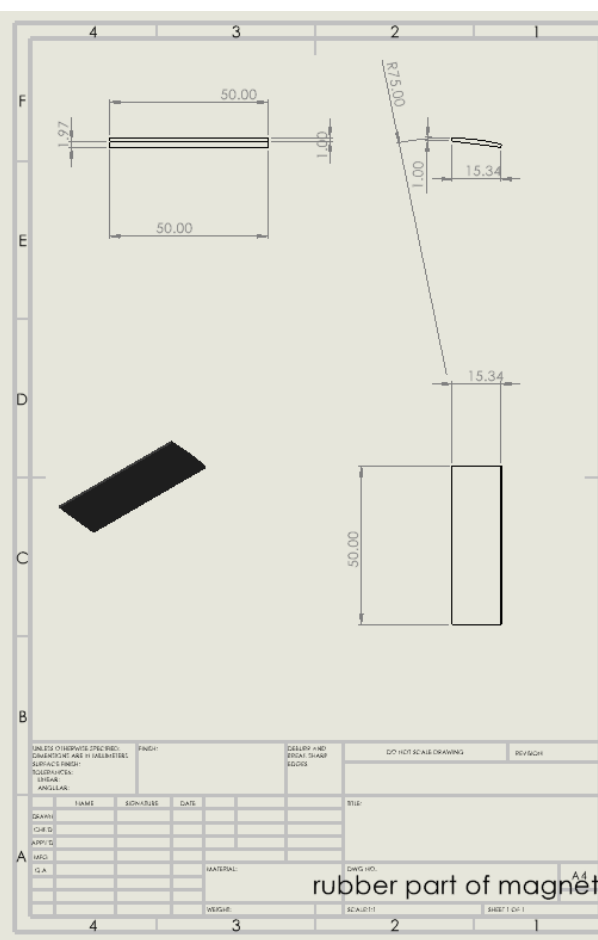
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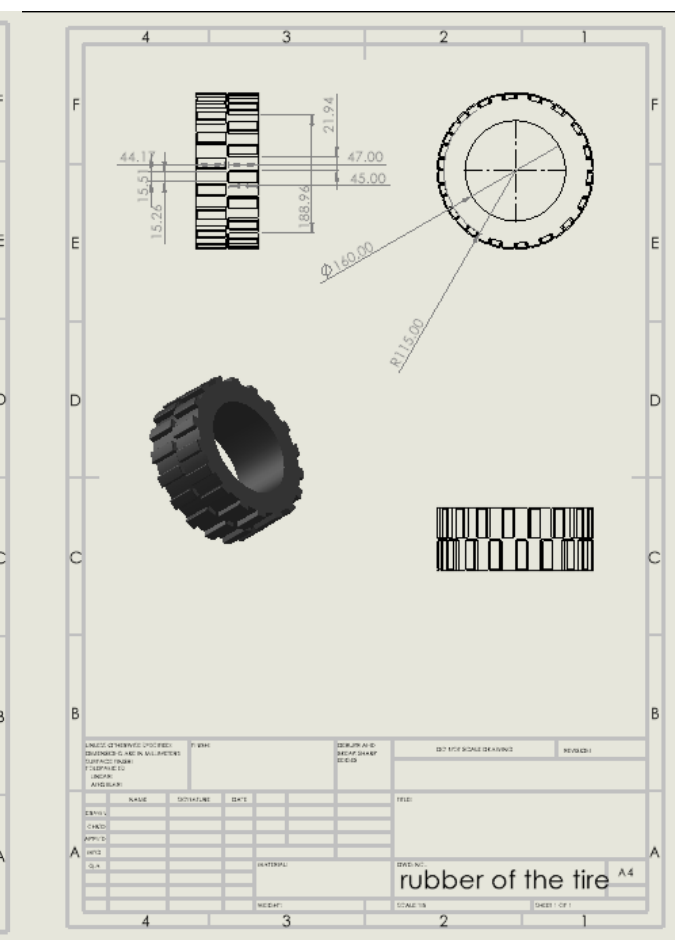
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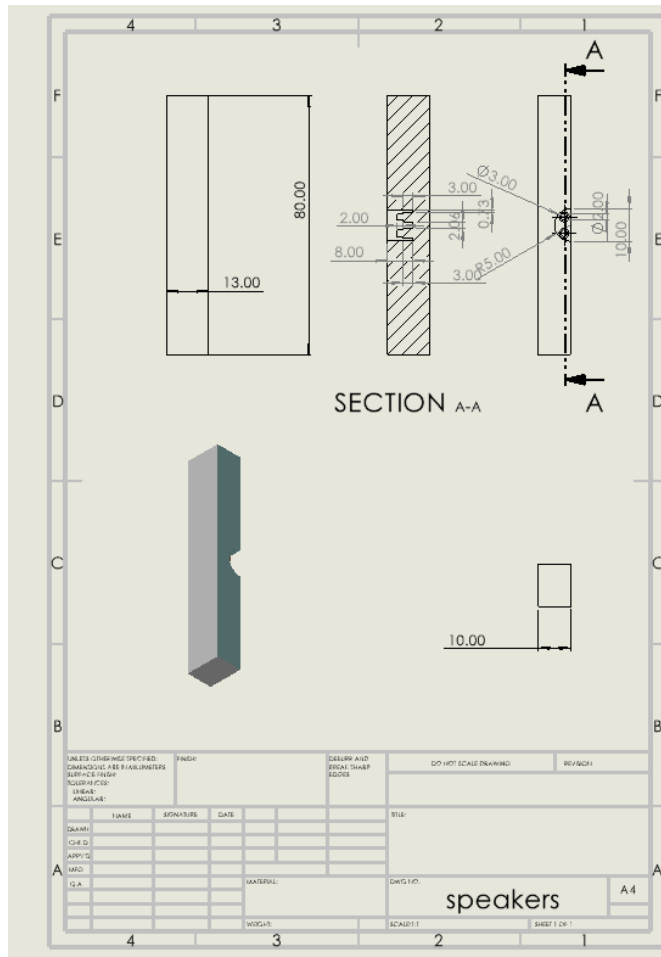
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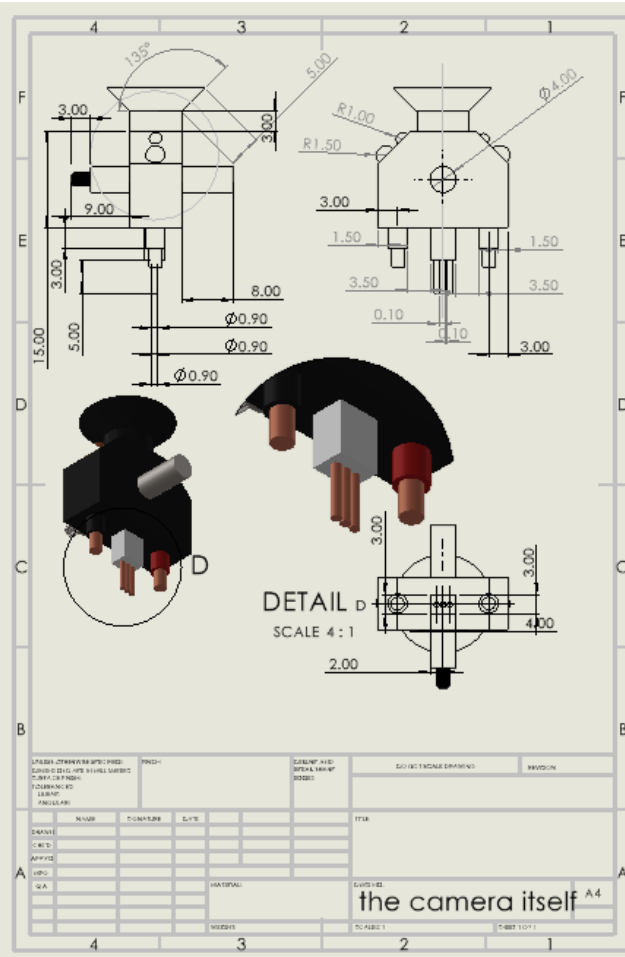
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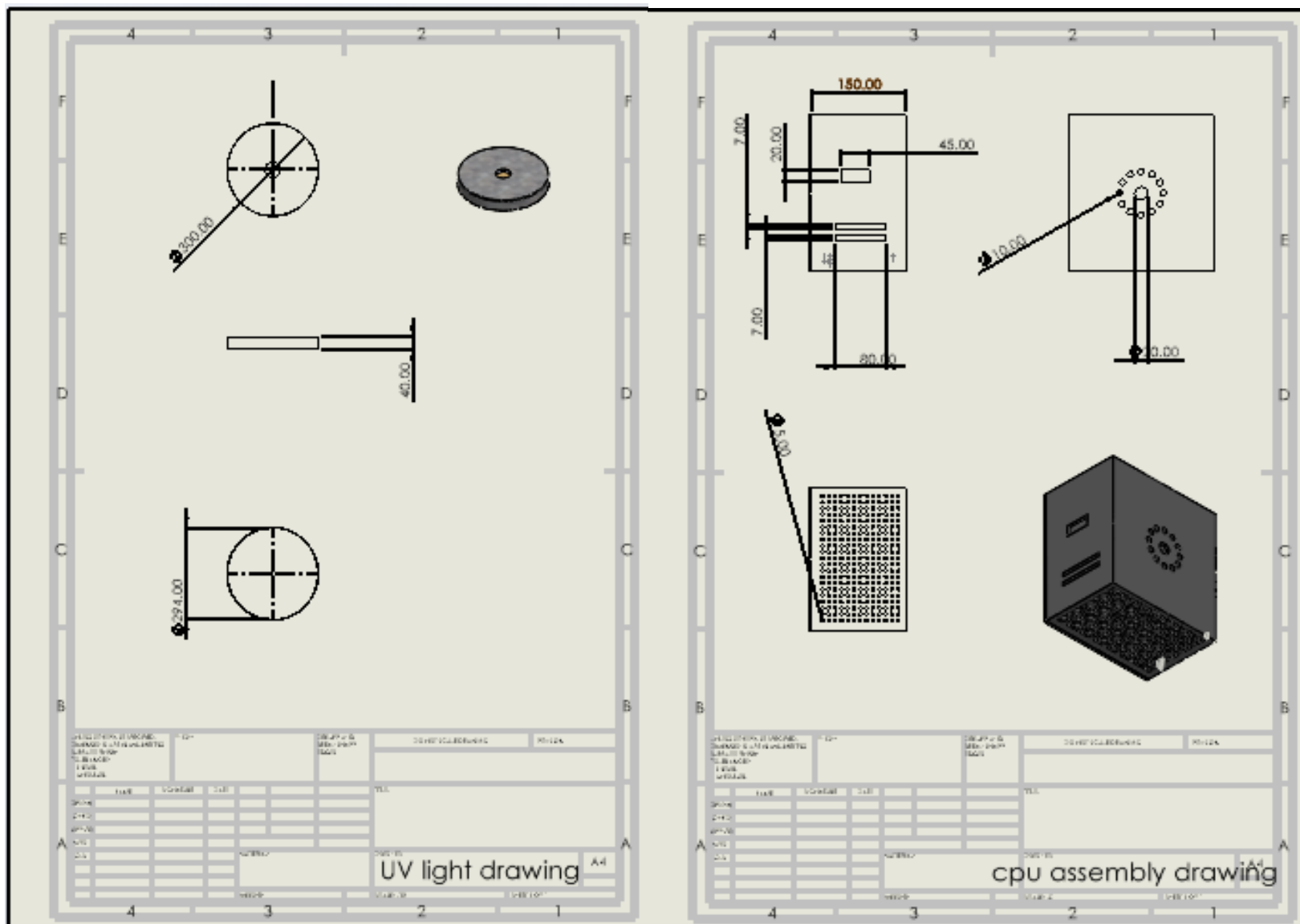


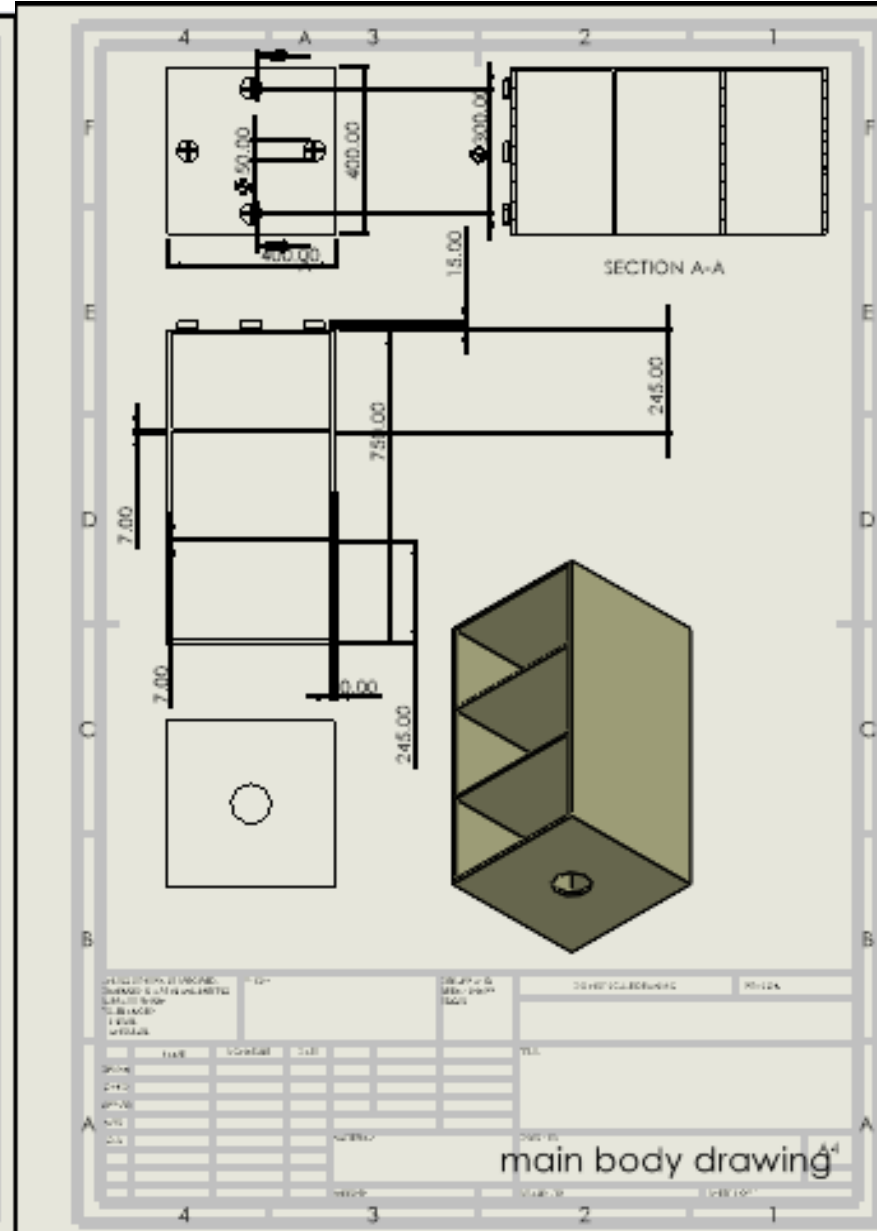
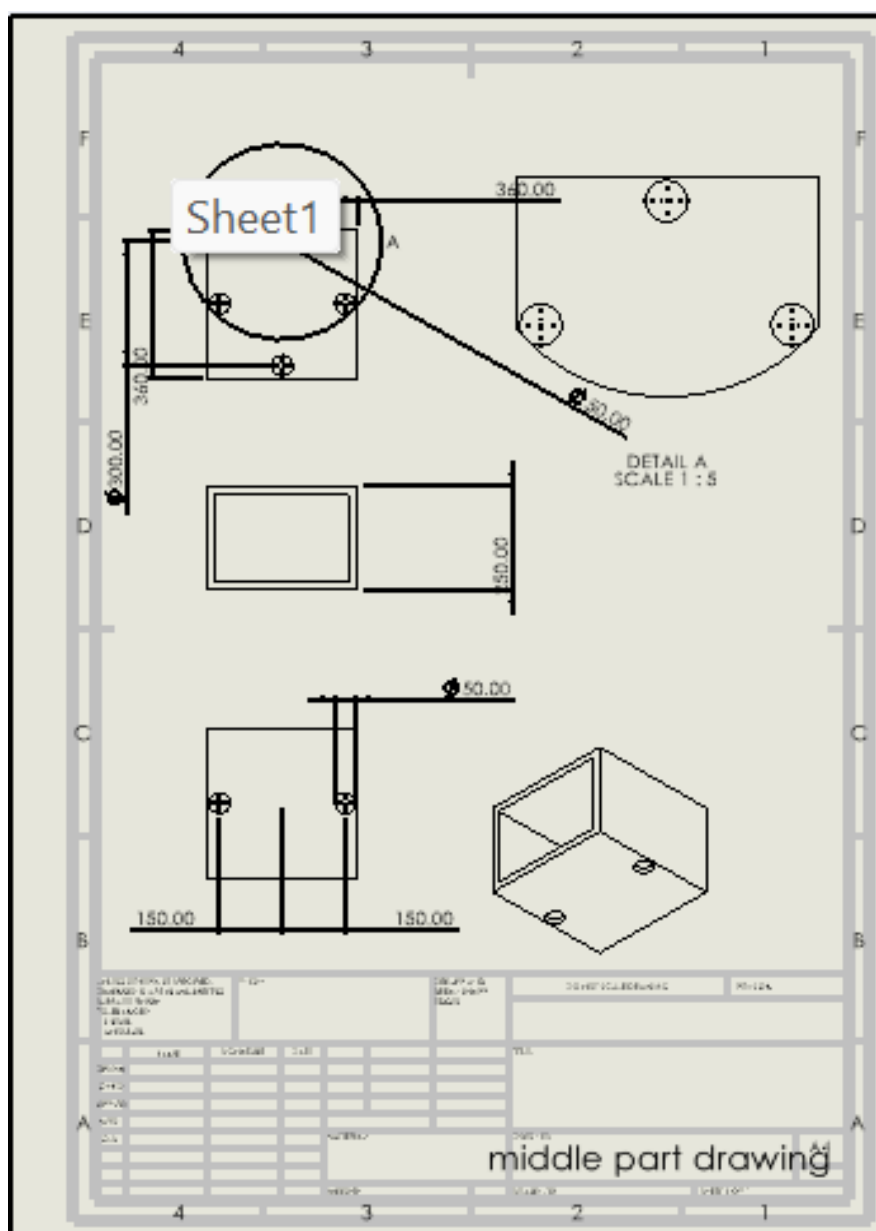
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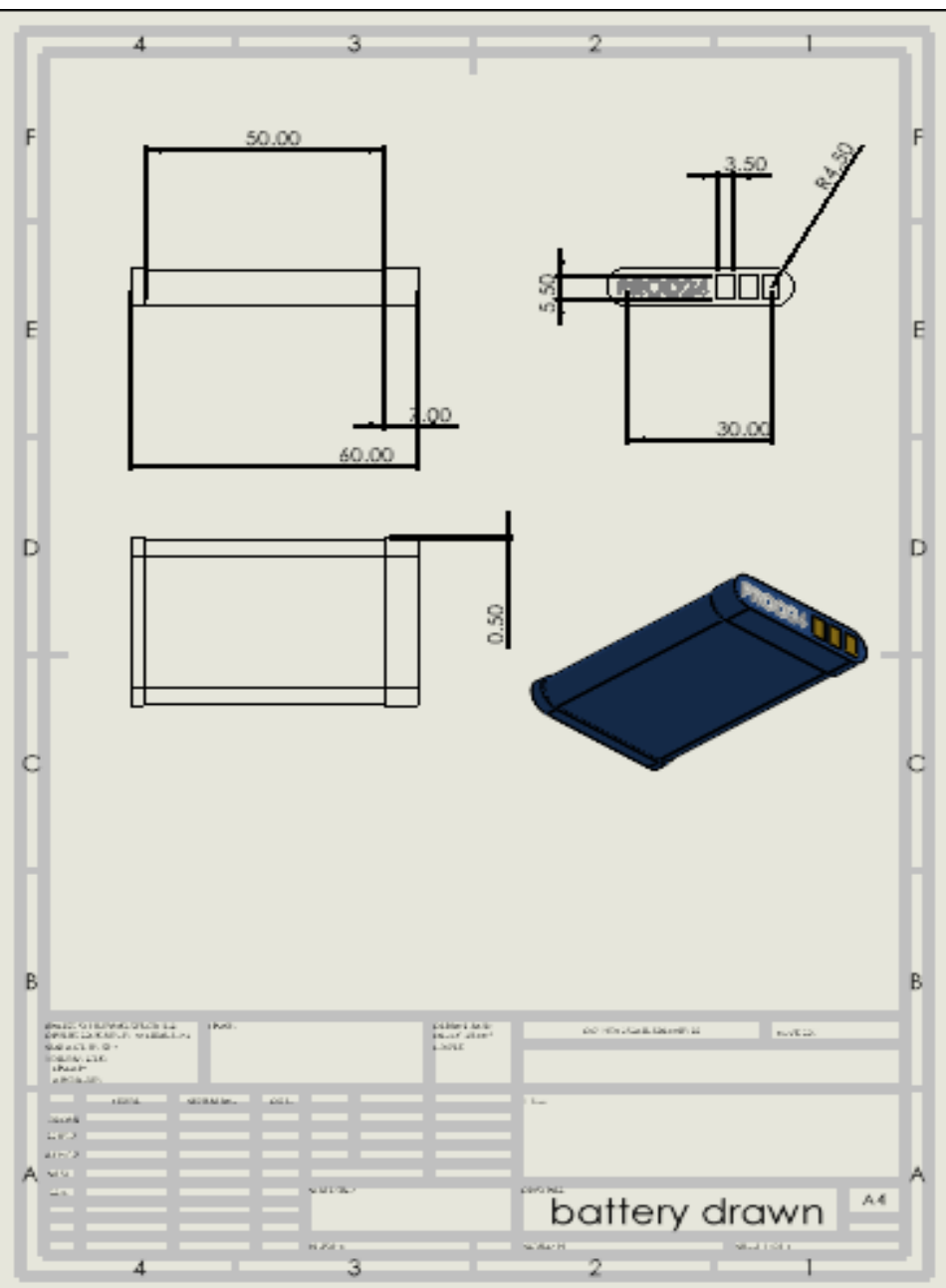
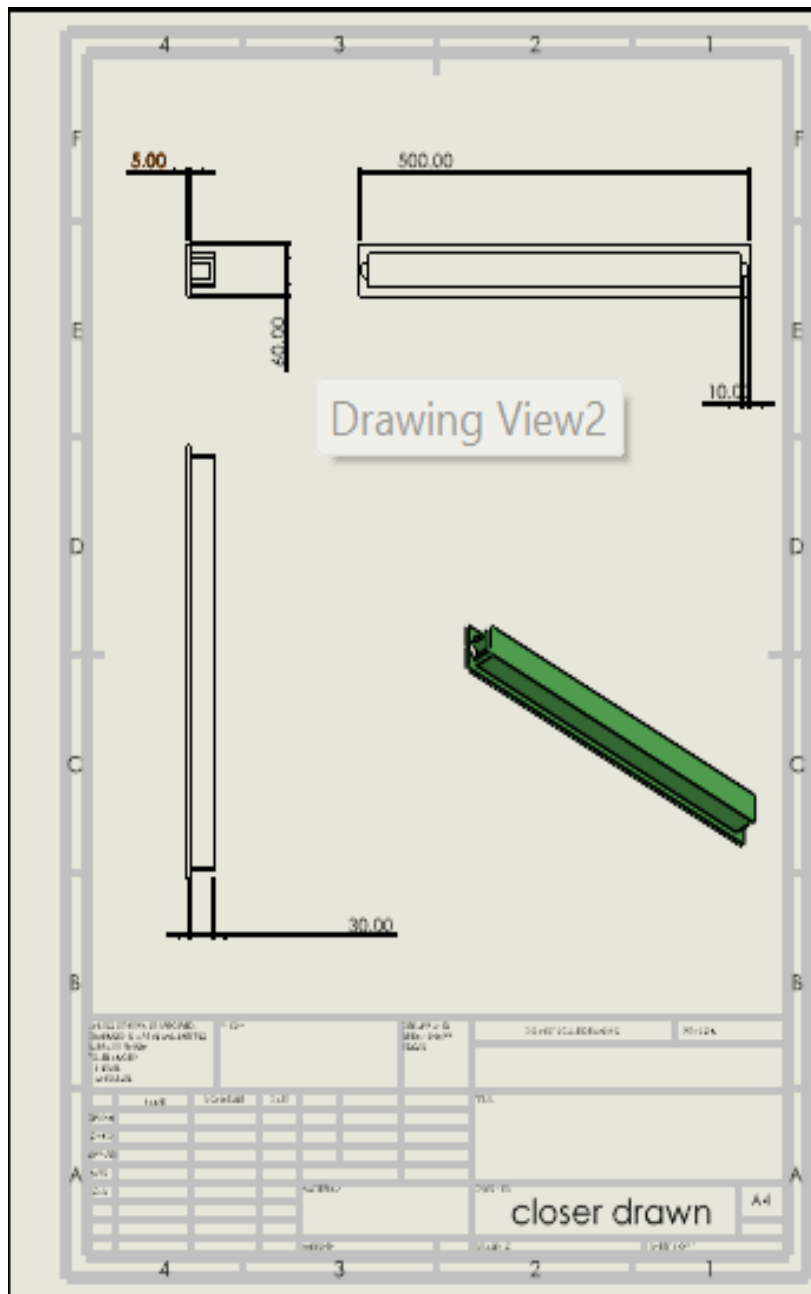


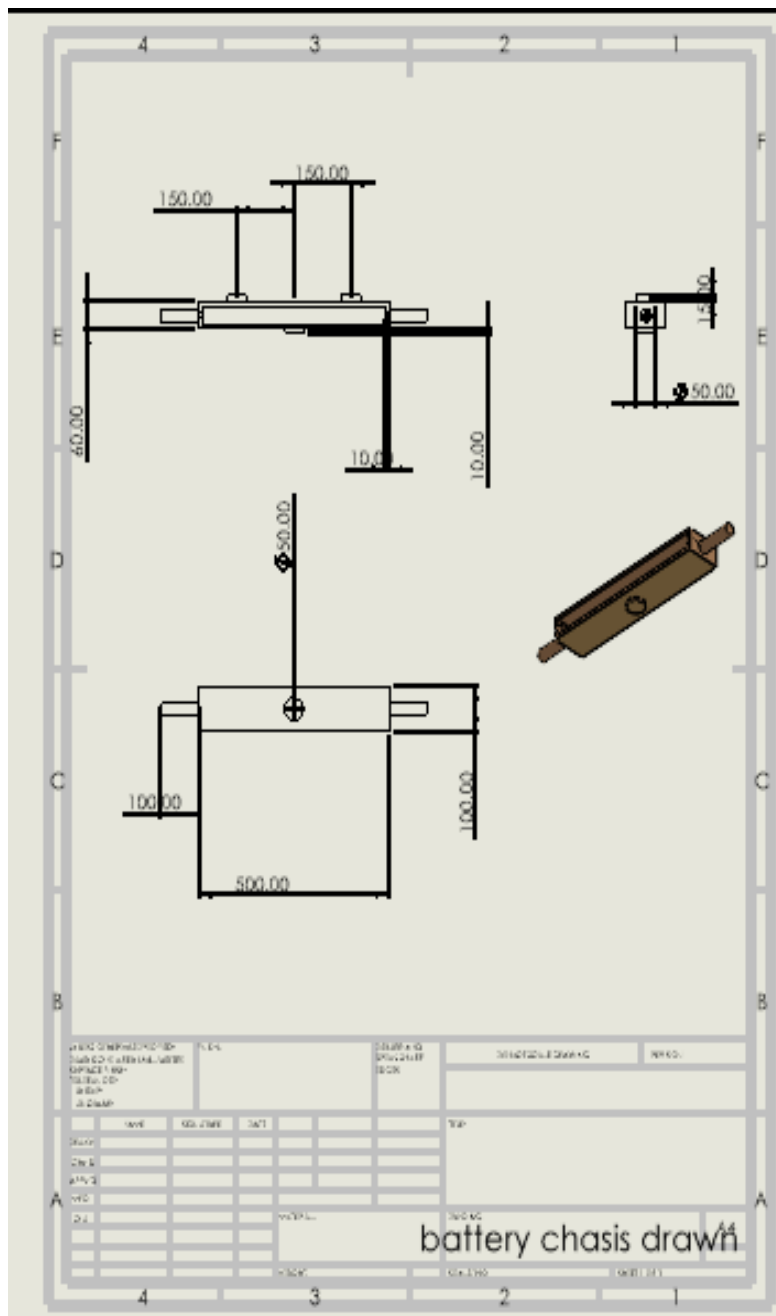
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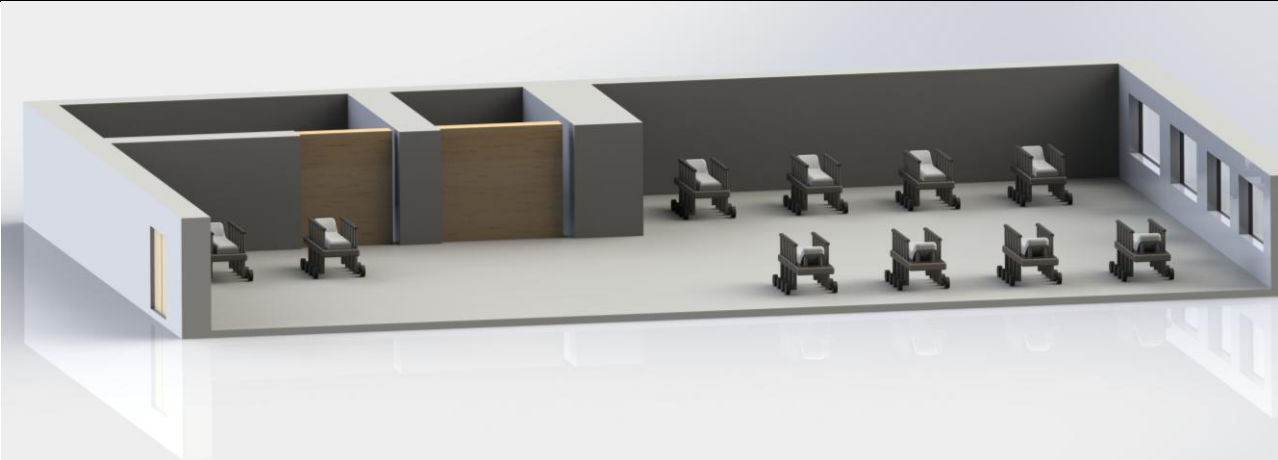


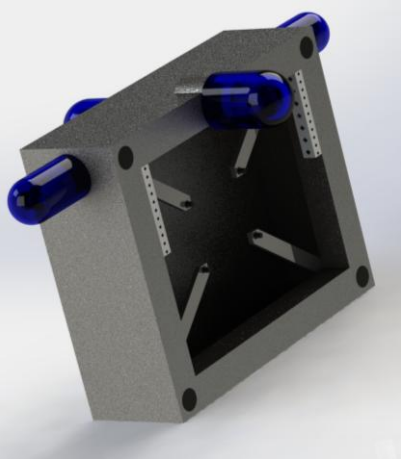
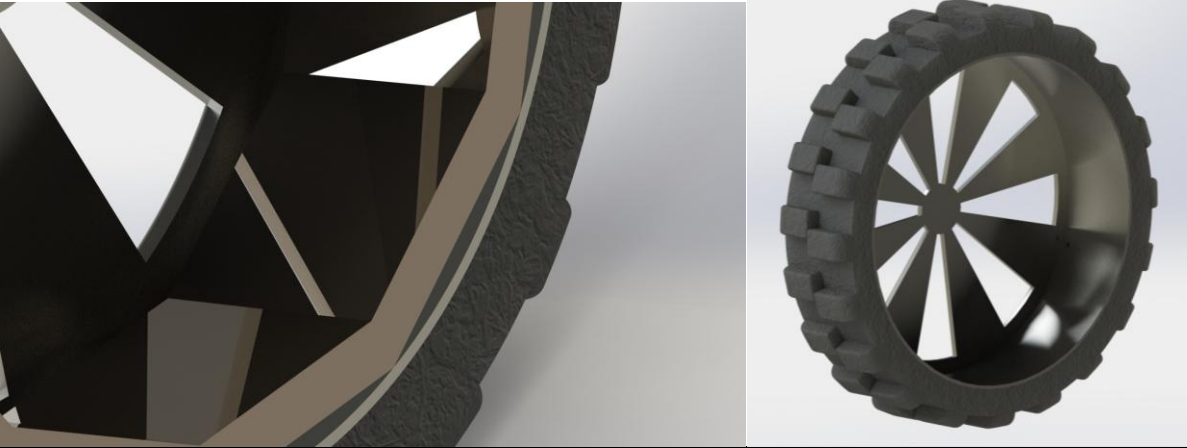
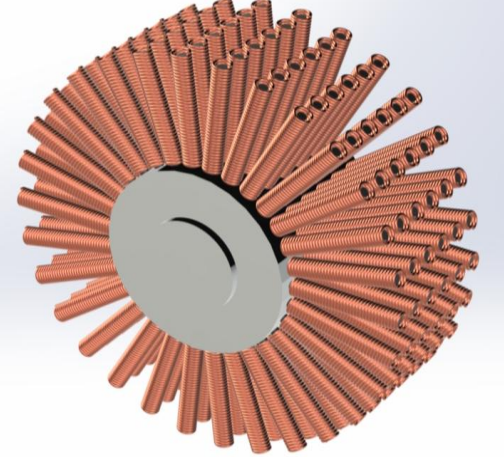




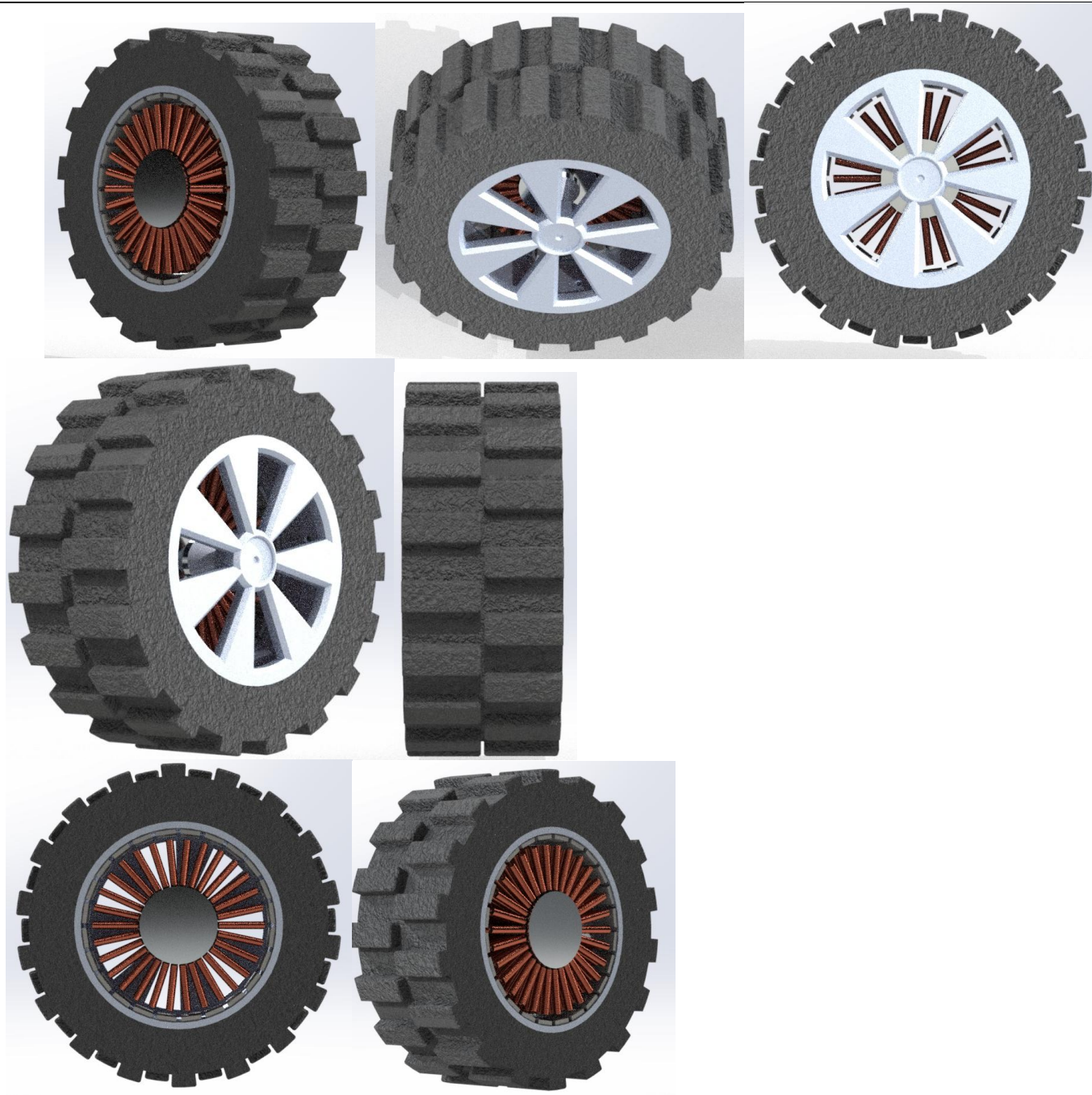




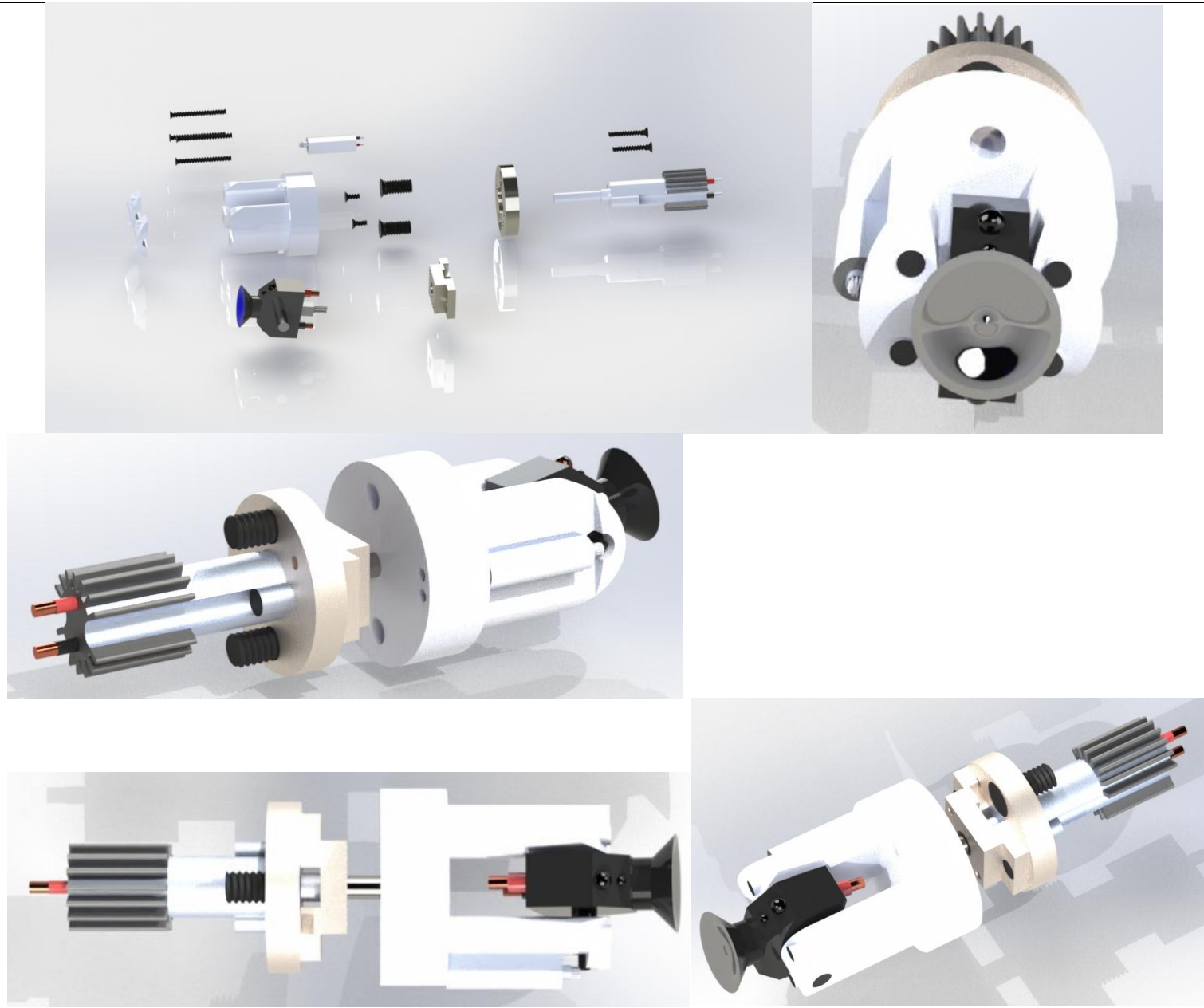
Item/ Assembly	Images
Hospital bed	
Hospital ward with bed	

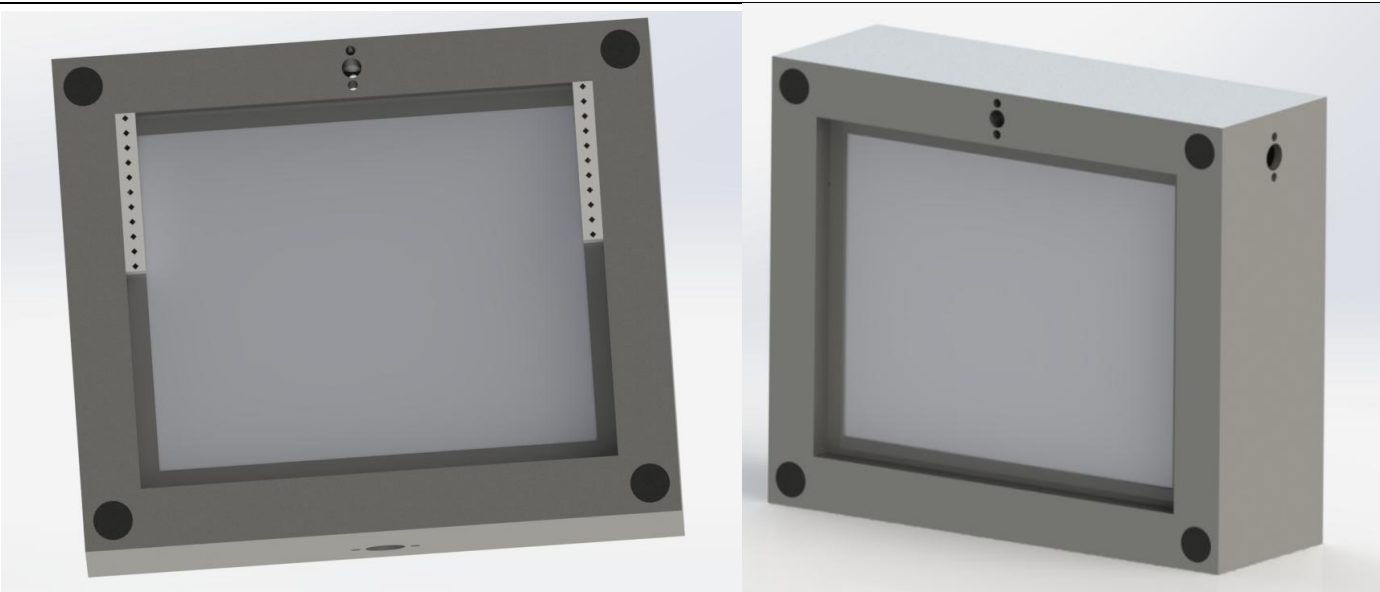
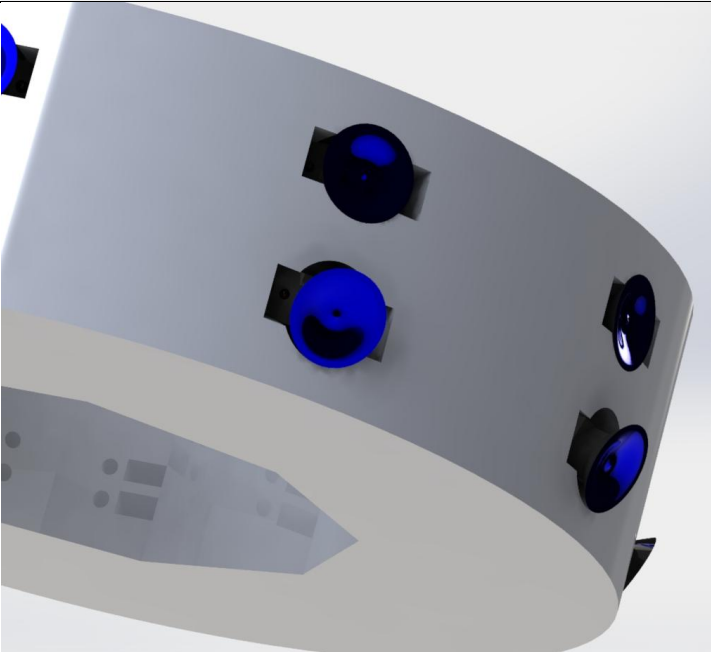
Head of the robot without screen	
Wheel without coils	
Coils for the wheel	

Wheel with coil

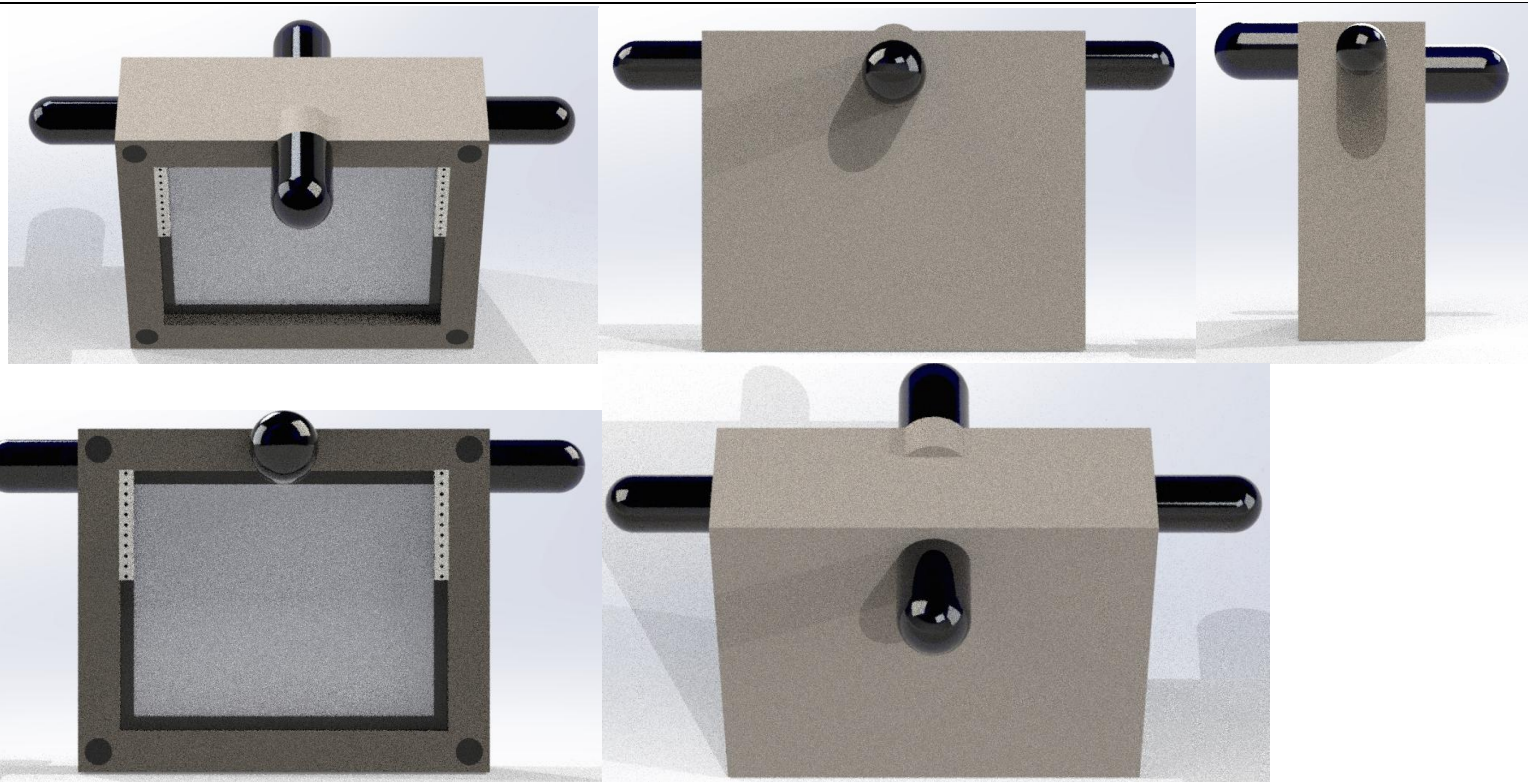


Camera module



Top part of robot without the cameras	
The circle of cameras	

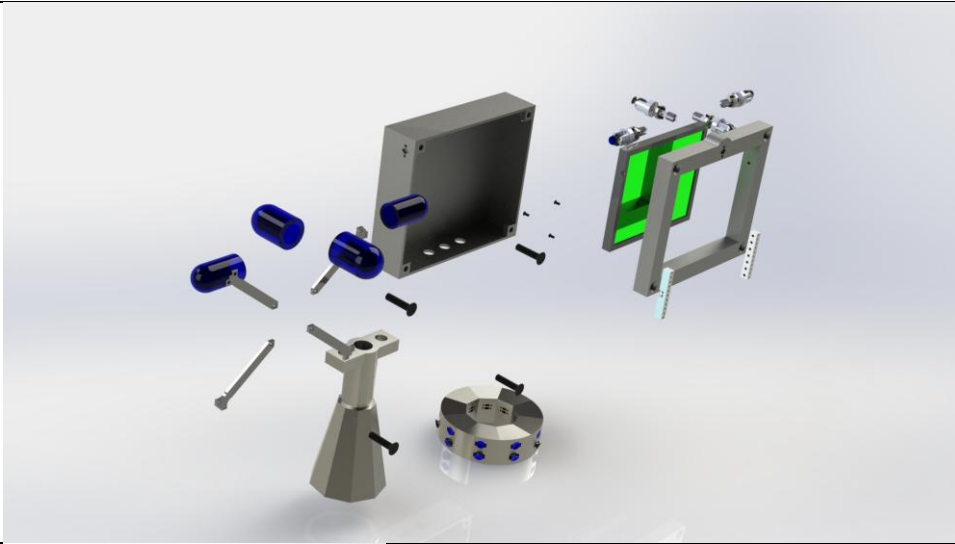
Top part of robot with cameras and protectors



Top part of robot with cameras and no protectors

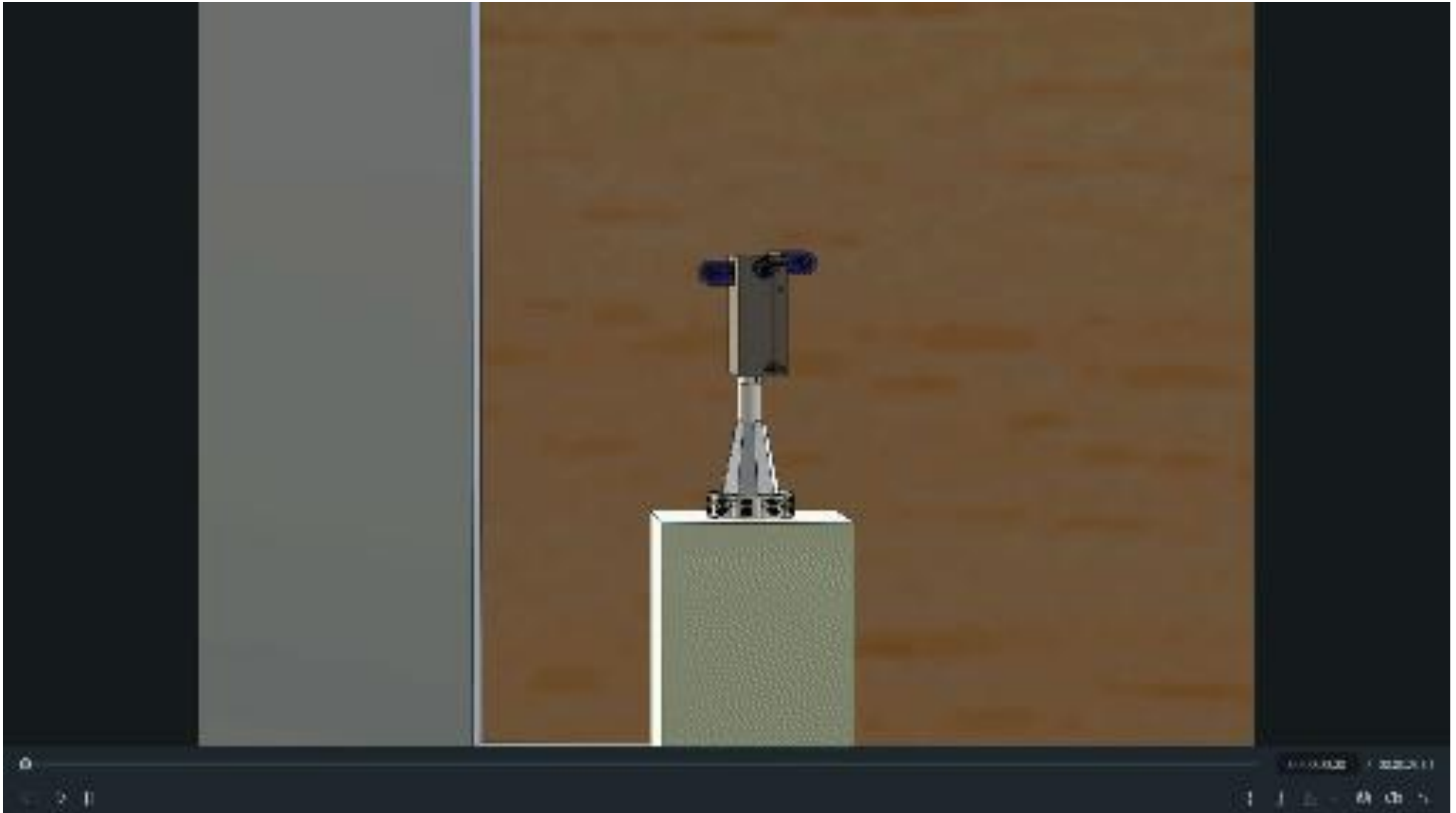


Exploded view of top part
and connections



Full robot





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Appendix: Part Selection

Camera	Microphone	Monitor	Speaker	Motor	Gyroscope	Processor
Resolution	Sensitivity	Size	Output range	Weight	Latency	Speed
Power Consumption	Response Range	Resolution	Latency	Efficiency	Format	Cost
Weight	Weight	Weight	Response	Power Consumption	Cost	Development Support
Cost	Cost	Cost	Cost	Cost	Power Consumption	Lifetime
Focal Distance	Power Consumption	Power Consumption	Weight	Lifetime	Axis Resolution	Power Consumption